



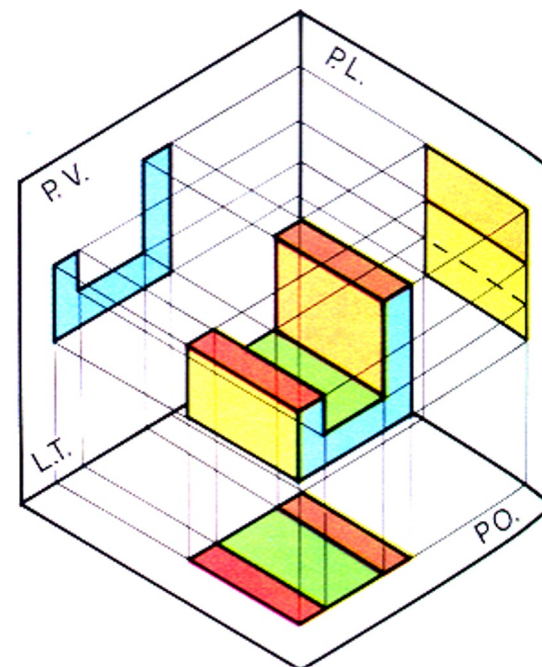
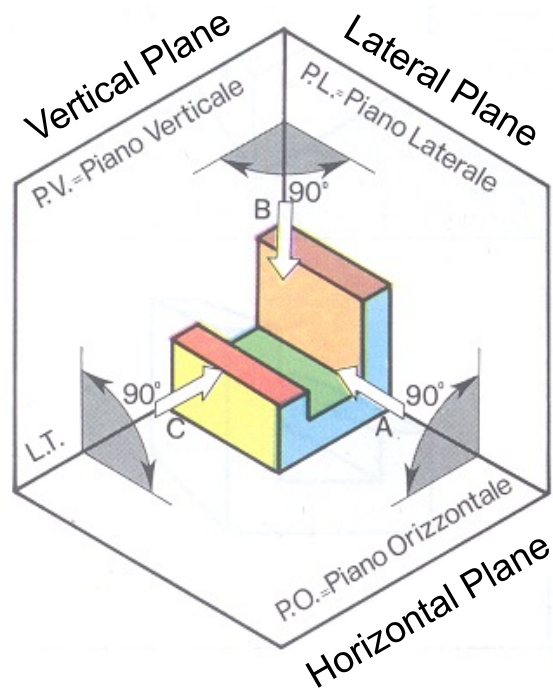
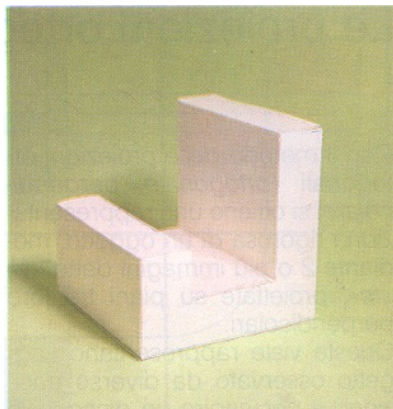
Geometrical Product Specifications

Standards: ISO 1101 and ASME Y14.5

Hocken R.J., P.H. Pereira (2012) Coordinate Measuring Machines and Systems, Chapters: 3 and 8

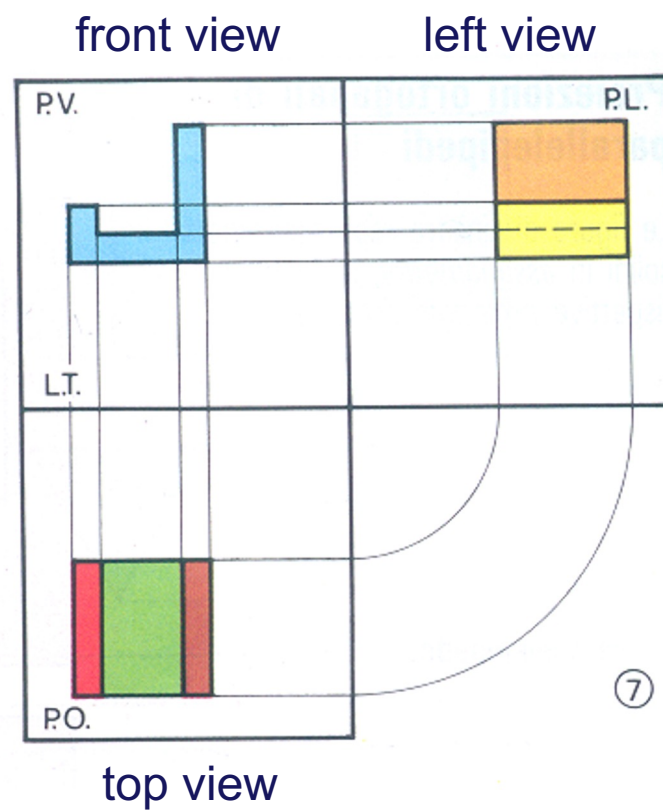
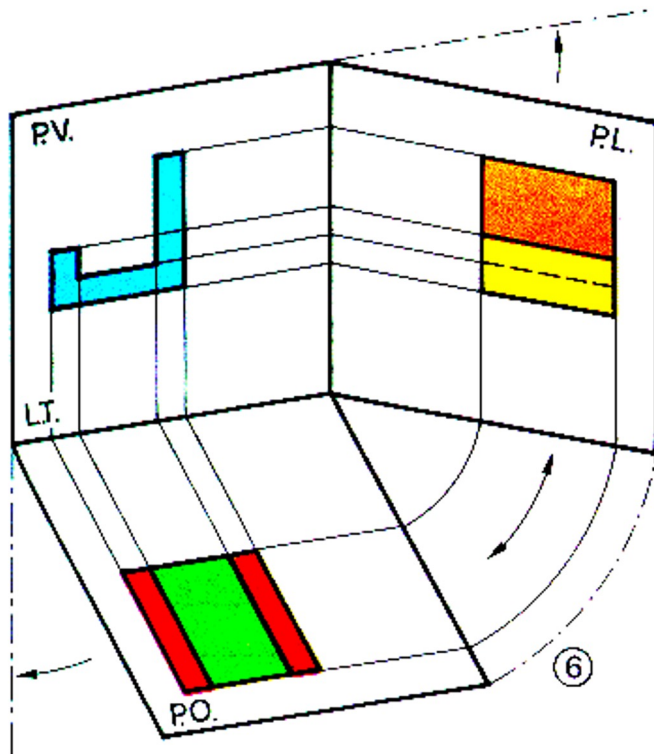


Multiview Orthographic Projection



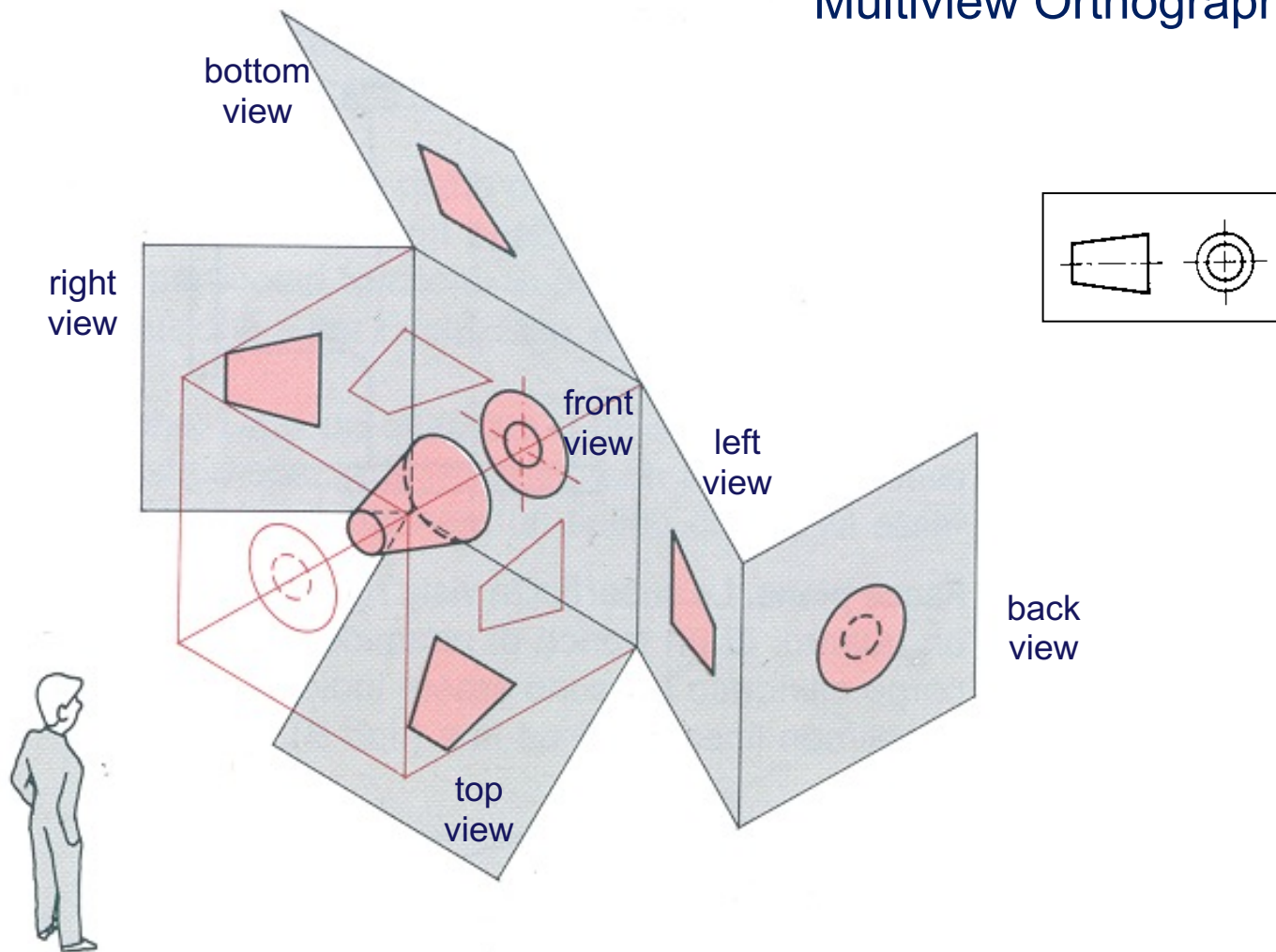


Multiview Orthographic Projection



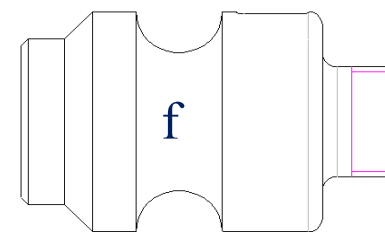
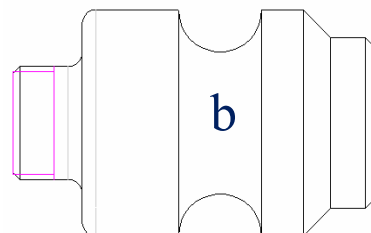
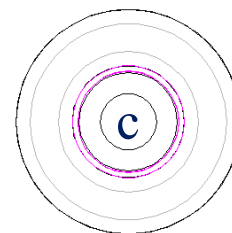
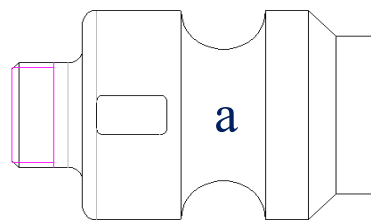
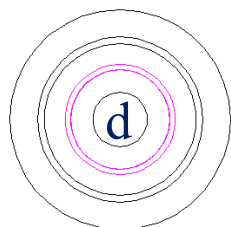
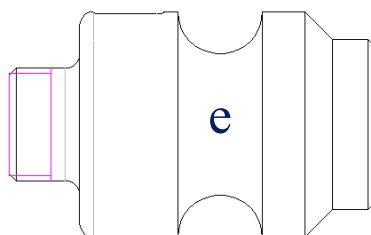
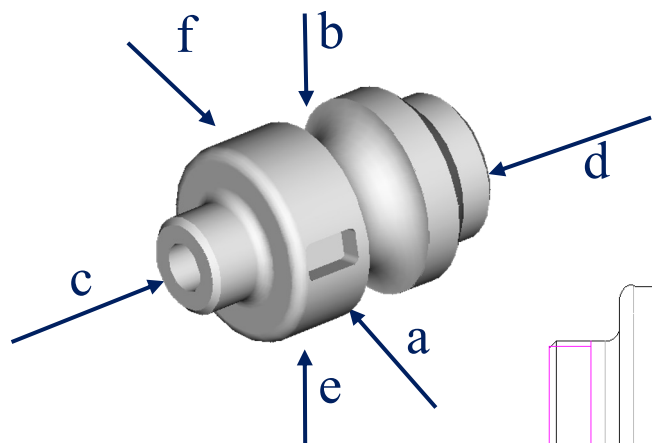


Multiview Orthographic Projection

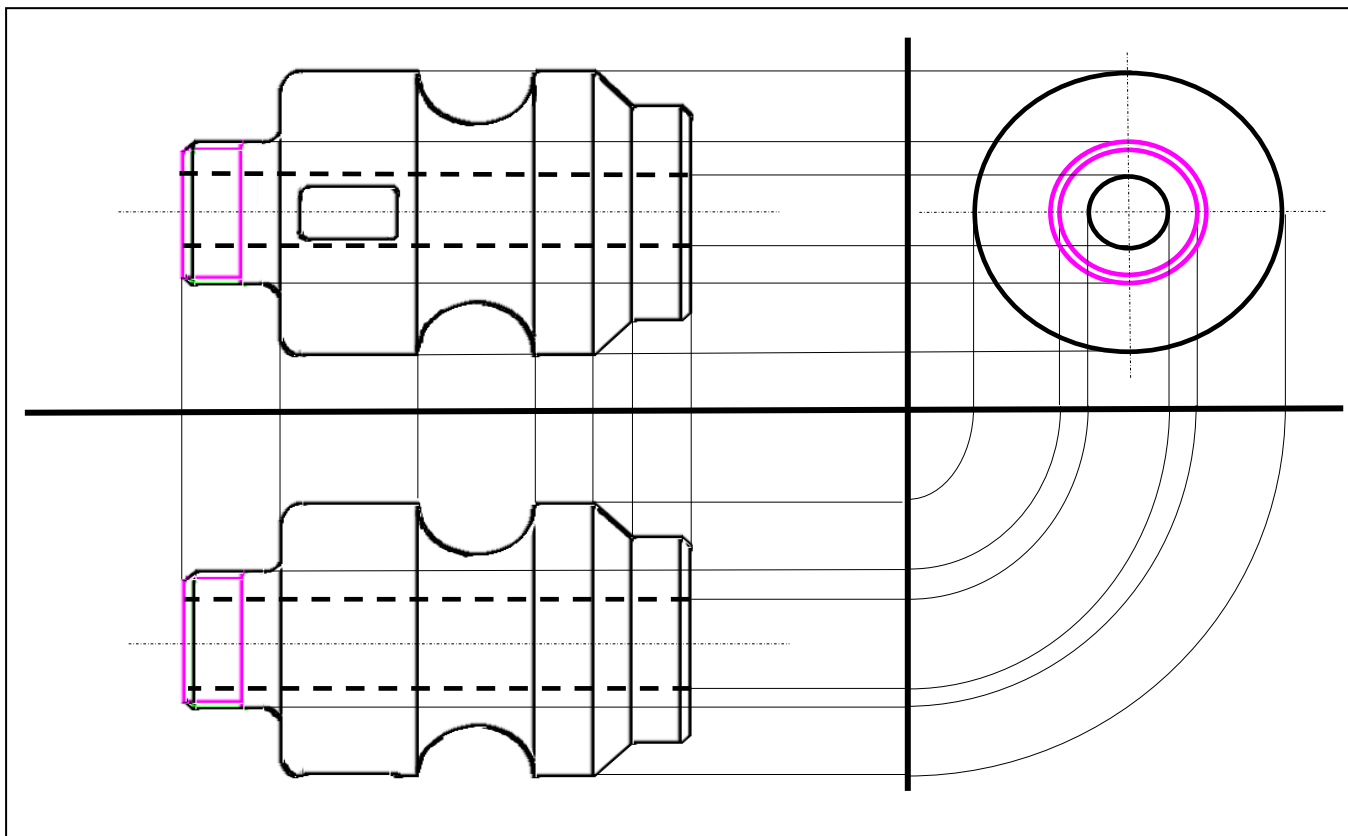


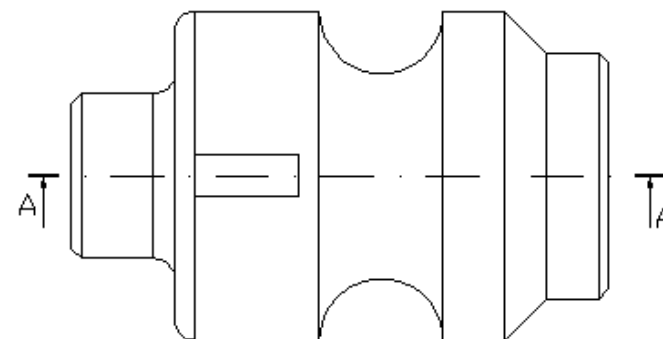
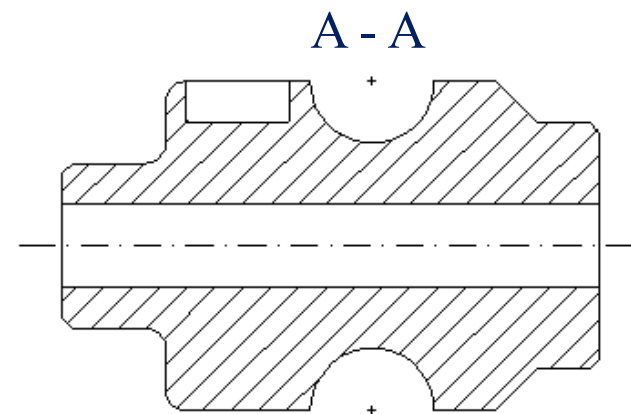
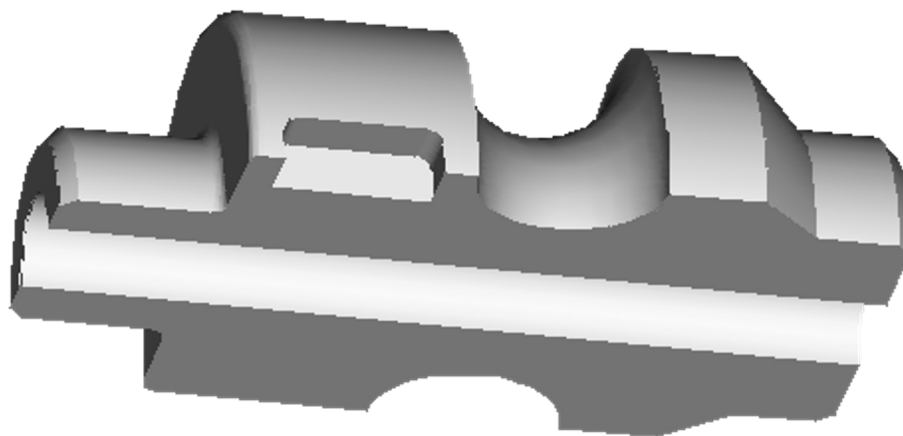


Multiview Orthographic Projection



- a. front view
- b. top view
- c. left view
- d. right view
- e. bottom view
- f. back view







GEOMETRICAL PRODUCT SPECIFICATIONS

Drawing specifications

- Set of lines, numerical values, and graphical indications used to define geometrical product specifications
- The goal is to assure that a product conforms to design specifications, from different points of view:
 - performance in use
 - fabrication and assembling
 - product verification

Factors that determine the performance of a manufactured product, other than mechanical and physical properties, include :

- **Dimensions** - linear or angular sizes specified on the part drawing
- **Tolerances** - allowable variations from the specified part dimensions that are permitted in manufacturing



DEFINITIONS (ANSI/ASME Y14.5M-2009)

A **dimension** is a numerical value expressed in appropriate units of measure and indicated on a drawing and in other documents along with lines, symbols, and notes to define the size or geometric characteristic, or both, of a part or part feature.

The dimension indicates the part size desired by the designer, if the part could be made with no errors or variations in the fabrication process.

A **tolerance** is the total amount by which a specific dimension is permitted to vary. The tolerance is the difference between the maximum and minimum limits.

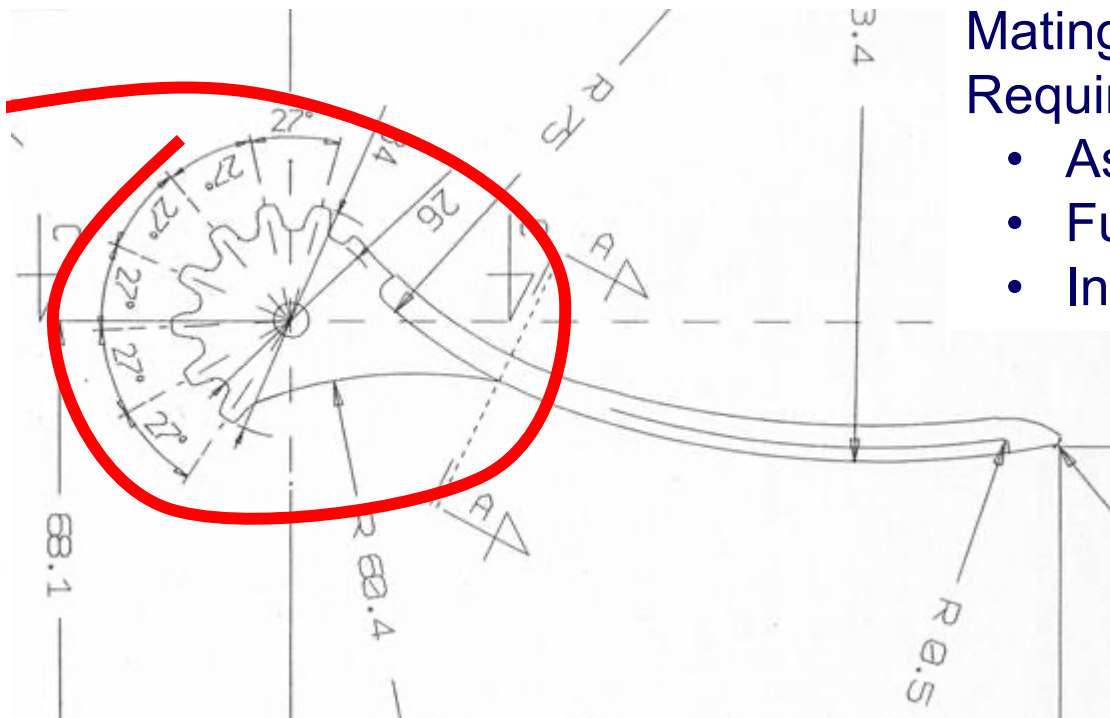
Macro-geometry: Size tolerances, Geometric tolerances

Micro-geometry: Waviness and Roughness



“Anna G.” 1994, design by Alessandro Medini
Corkscrew in chromed zamac and polyamide
NB: ZAMAC = Zinc-Aluminum-Magnesium-Copper

Technical drawing of the leverage

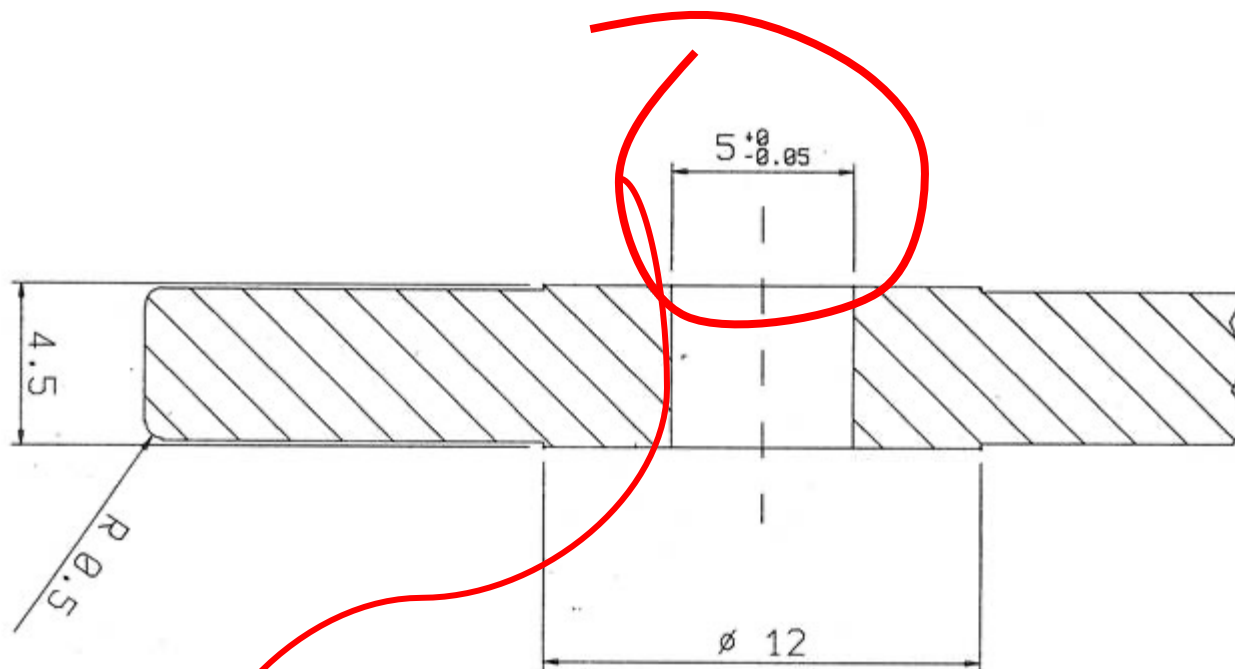


Mating pin-leverage
Requirements:

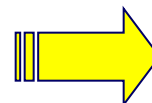
- Assemblability;
- Functionality;
- Interchangeability.



Mating pin - leverage



Diameter: $5 \begin{smallmatrix} +0 \\ -0,05 \end{smallmatrix}$

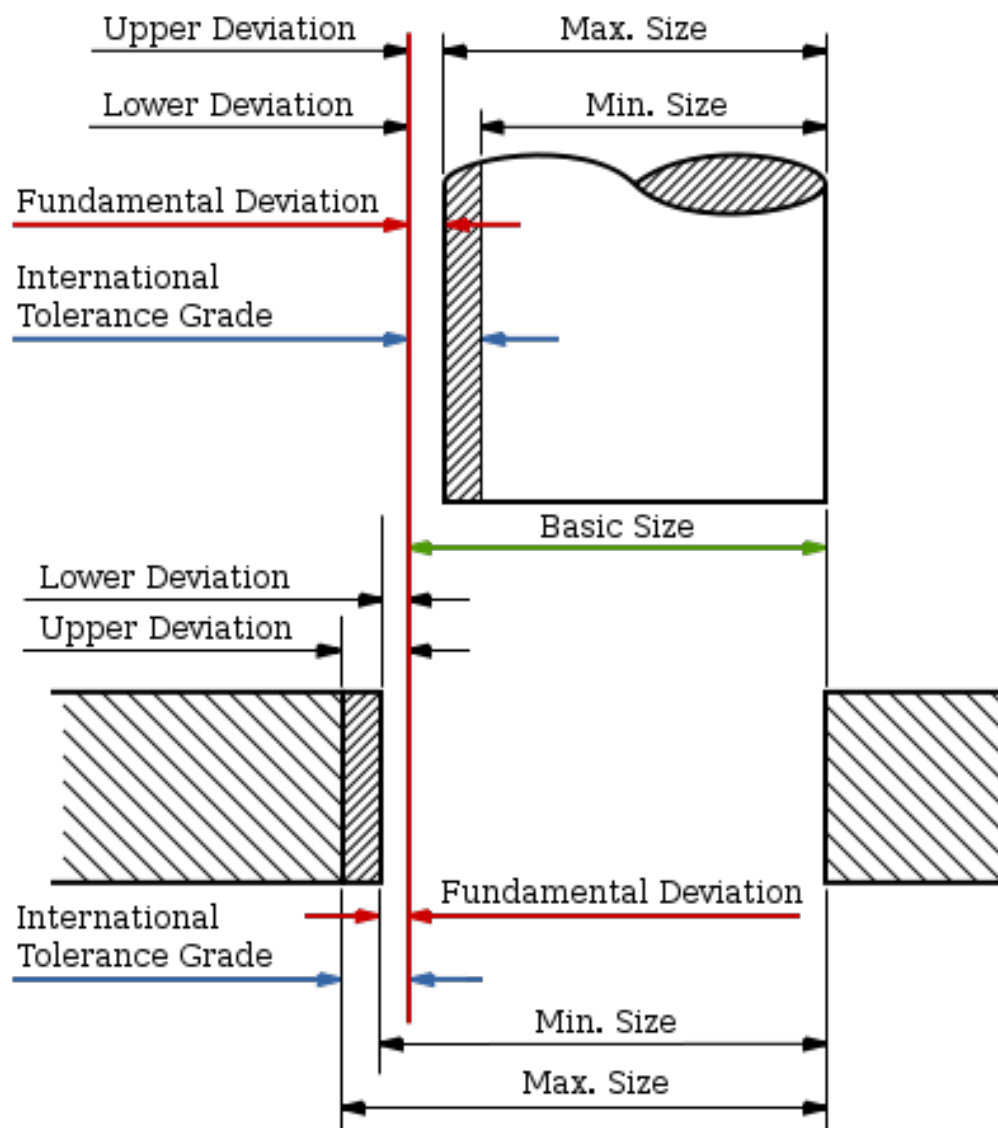


$4,95 \div 5$



SIZE TOLERANCES

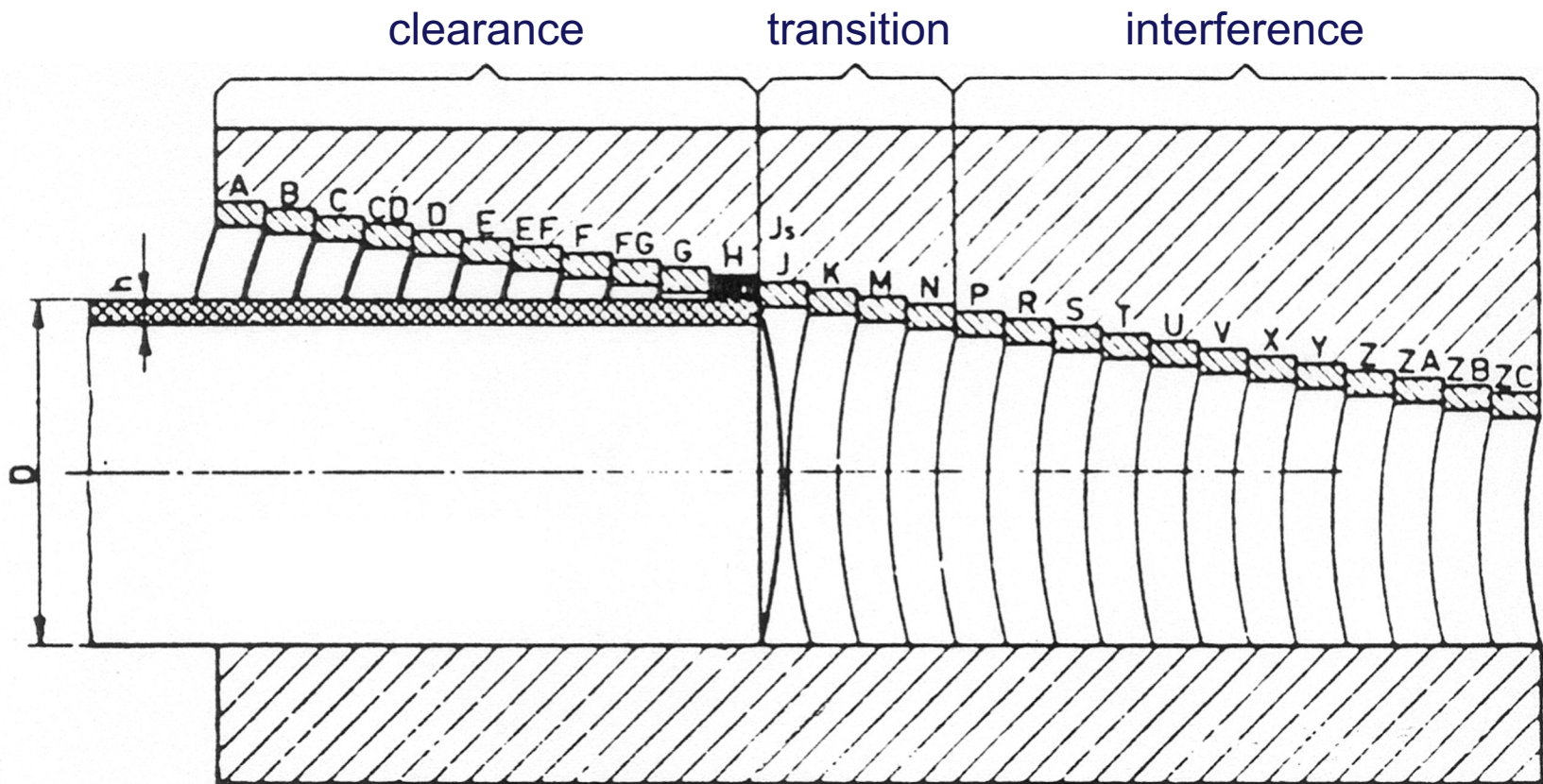
12





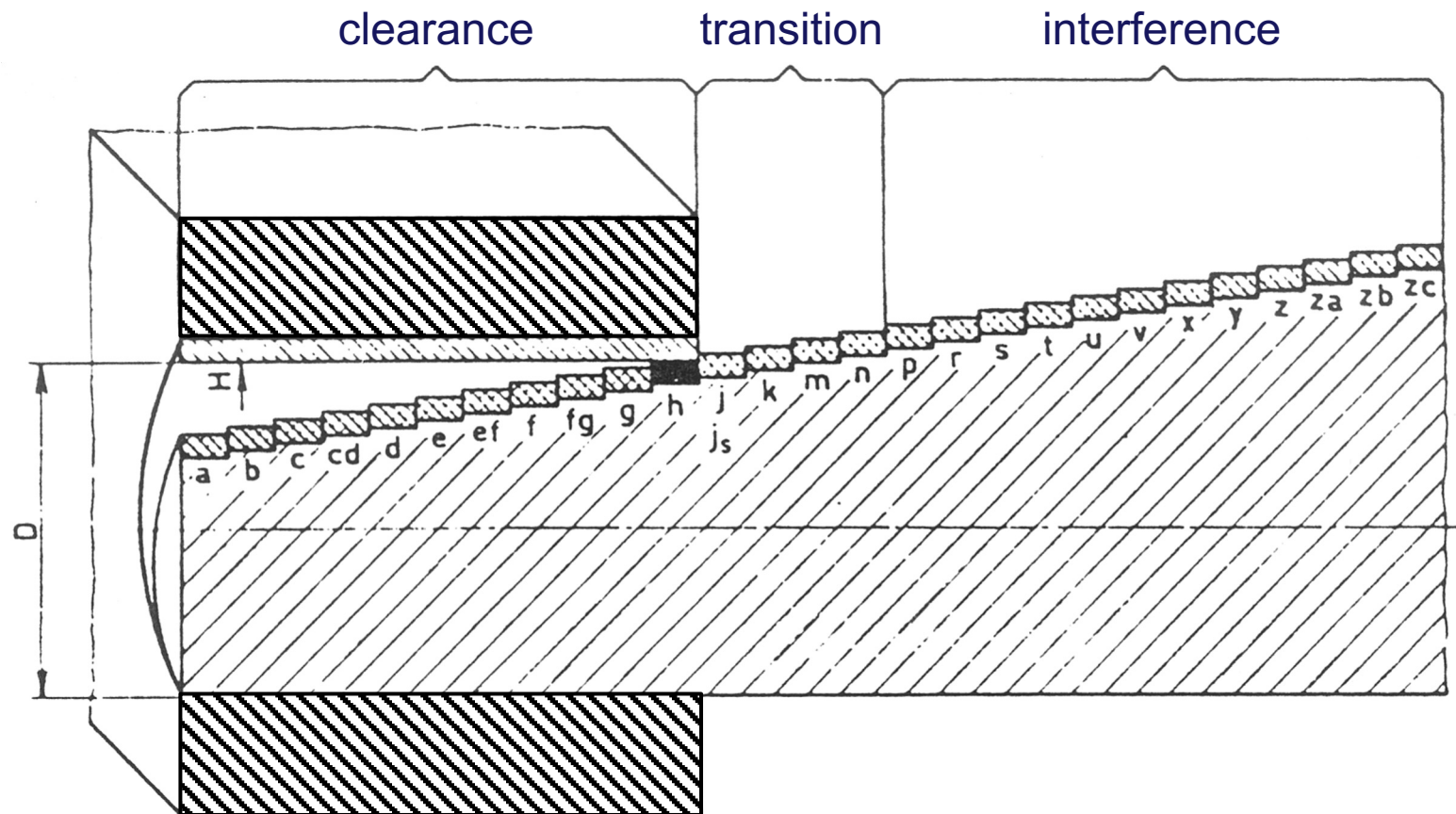
ISO SYSTEM OF LIMITS AND FITS (ISO 286)

Shaft basis system: size of the shaft is kept constant, hole size is varied to get different fits





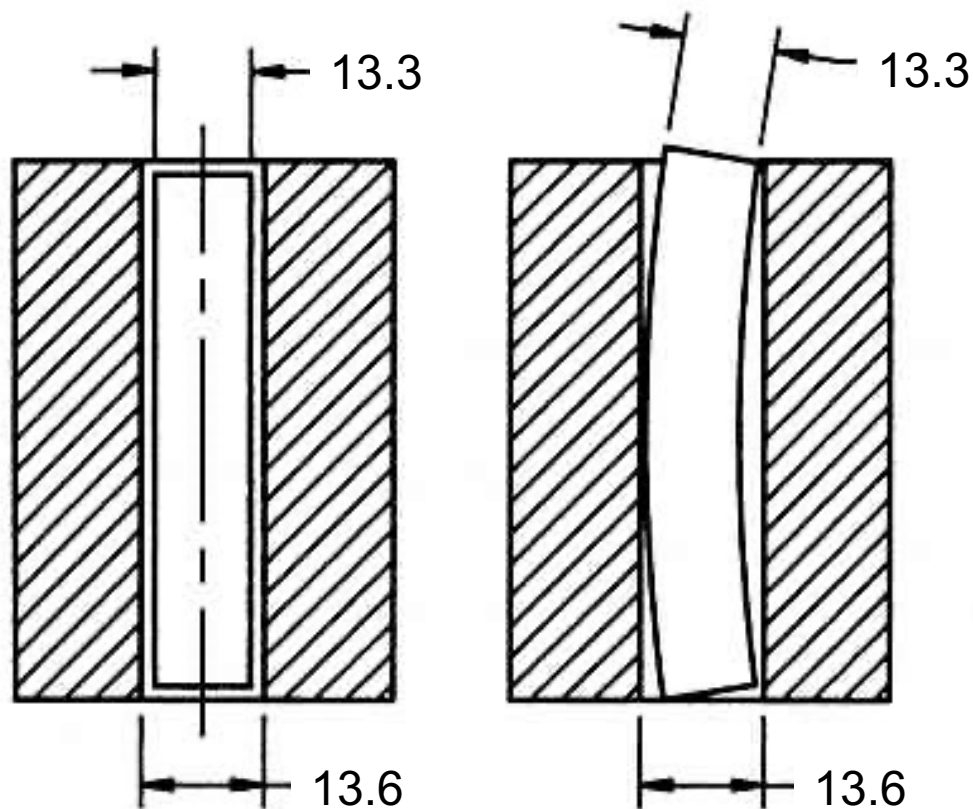
Hole basis system: size of the hole is kept constant, shaft size is varied to get different fits





INTERNATIONAL TOLERANCE GRADE (ISO 286)

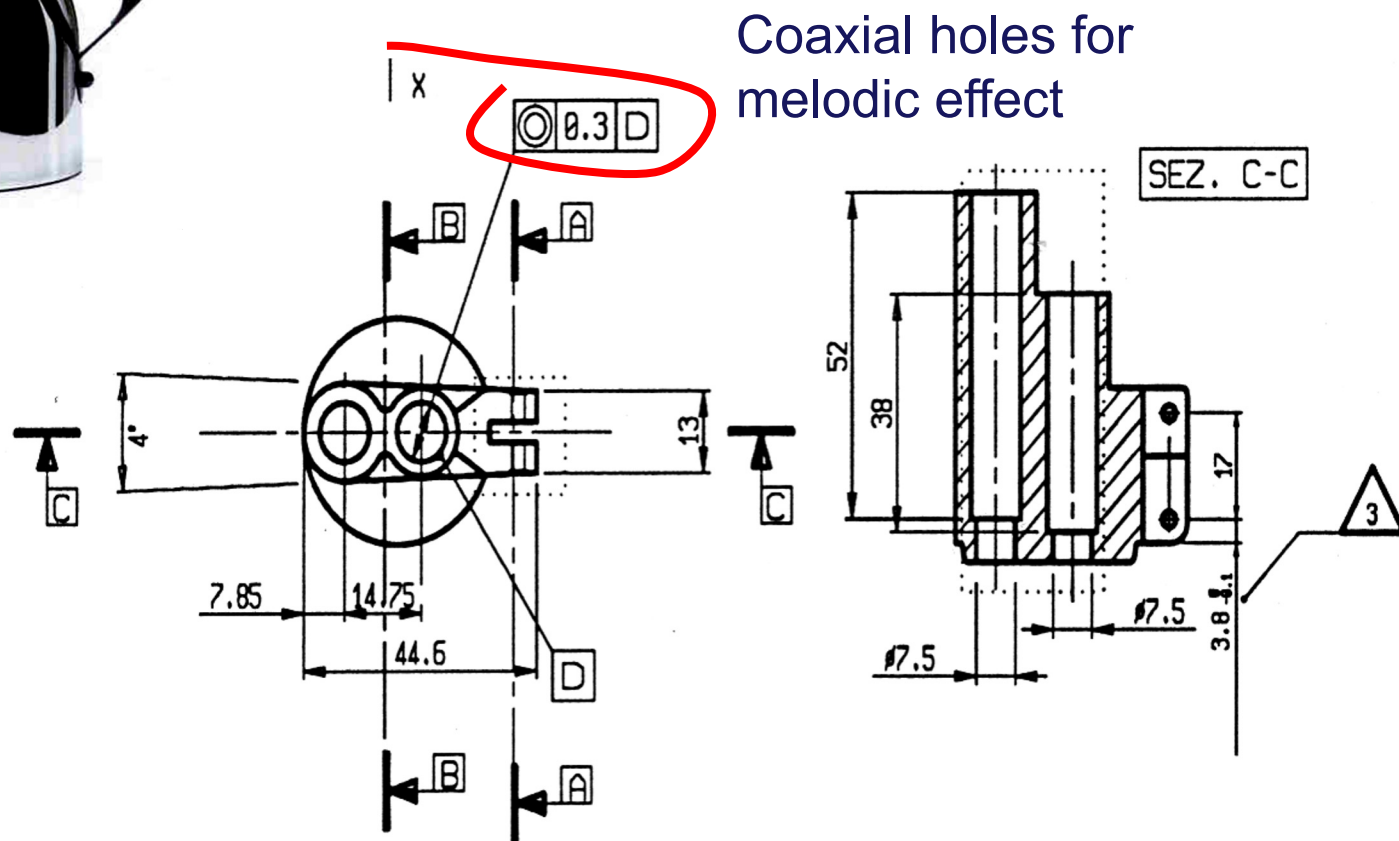
Basic size mm		Standard tolerance grades																	
		IT1 ²⁾	IT2 ²⁾	IT3 ²⁾	IT4 ²⁾	IT5 ²⁾	IT6	IT7	IT8	IT9	IT10	IT11	IT12	IT13	IT14 ³⁾	IT15 ³⁾	IT16 ³⁾	IT17 ³⁾	IT18 ³⁾
Above	Up to and including	Tolerances																	
		μm											mm						
—	3 ³⁾	0,8	1,2	2	3	4	6	10	14	25	40	60	0,1	0,14	0,25	0,4	0,6	1	1,4
3	6	1	1,5	2,5	4	5	8	12	18	30	48	75	0,12	0,18	0,3	0,48	0,75	1,2	1,8
6	10	1	1,5	2,5	4	6	9	15	22	36	58	90	0,15	0,22	0,36	0,58	0,9	1,5	2,2
10	18	1,2	2	3	5	8	11	18	27	43	70	110	0,18	0,27	0,43	0,7	1,1	1,8	2,7
18	30	1,5	2,5	4	6	9	13	21	33	52	84	130	0,21	0,33	0,52	0,84	1,3	2,1	3,3
30	50	1,5	2,5	4	7	11	16	25	39	62	100	160	0,25	0,39	0,62	1	1,6	2,5	3,9
50	80	2	3	5	8	13	19	30	46	74	120	190	0,3	0,46	0,74	1,2	1,9	3	4,6
80	120	2,5	4	6	10	15	22	35	54	87	140	220	0,35	0,54	0,87	1,4	2,2	3,5	5,4
120	180	3,5	5	8	12	18	25	40	63	100	160	250	0,4	0,63	1	1,6	2,5	4	6,3
180	250	4,5	7	10	14	20	29	46	72	115	185	290	0,46	0,72	1,15	1,85	2,9	4,6	7,2
250	315	6	8	12	16	23	32	52	81	130	210	320	0,52	0,81	1,3	2,1	3,2	5,2	8,1
315	400	7	9	13	18	25	36	57	89	140	230	360	0,57	0,89	1,4	2,3	3,6	5,7	8,9
400	500	8	10	15	20	27	40	63	97	155	250	400	0,63	0,97	1,55	2,5	4	6,3	9,7
500	630 ²⁾	9	11	16	22	32	44	70	110	175	280	440	0,7	1,1	1,75	2,8	4,4	7	11
630	800 ²⁾	10	13	18	25	36	50	80	125	200	320	500	0,8	1,25	2	3,2	5	8	12,5
800	1000 ²⁾	11	15	21	28	40	56	90	140	230	360	560	0,9	1,4	2,3	3,6	5,6	9	14
1000	1250 ²⁾	13	18	24	33	47	66	105	165	260	420	660	1,05	1,65	2,6	4,2	6,6	10,5	16,5
1250	1600 ²⁾	15	21	29	39	55	78	125	195	310	500	780	1,25	1,95	3,1	5	7,8	12,5	19,5
1600	2000 ²⁾	18	25	35	46	65	92	150	230	370	600	920	1,5	2,3	3,7	6	9,2	15	23
2000	2500 ²⁾	22	30	41	55	78	110	175	280	440	700	1100	1,75	2,8	4,4	7	11	17,5	28
2500	3150 ²⁾	26	36	50	68	96	135	210	330	540	860	1350	2,1	3,3	5,4	8,6	13,5	21	33



Size tolerances are often not enough!

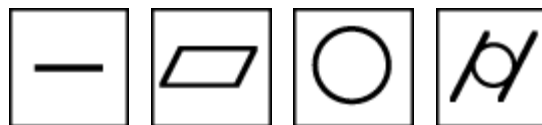


“9091” - 1985
design by Richard Sapper
Kettle with melodic whistle

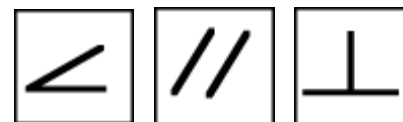




Form



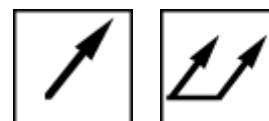
Orientation



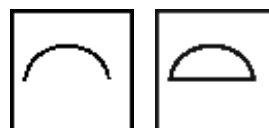
Location



Run-out



Profile



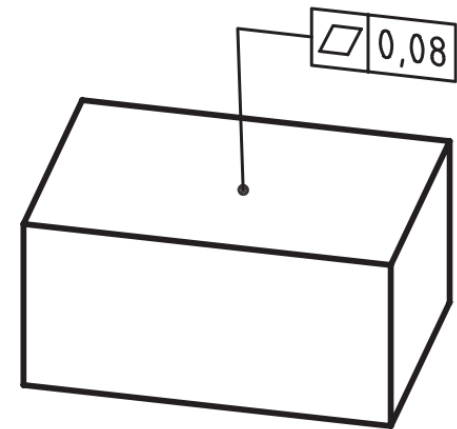
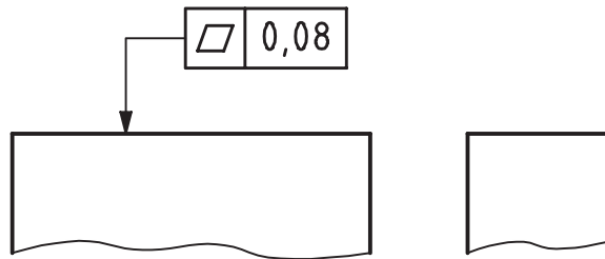


FORM TOLERANCES: FLATNESS

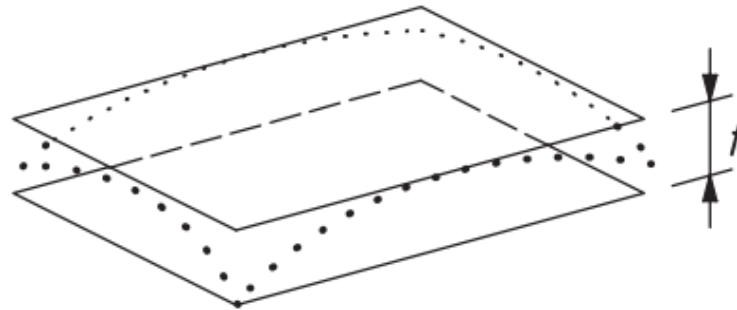
Applicability:

- Integral flat surfaces
- Derived features (median planes)

Indication



Interpretation

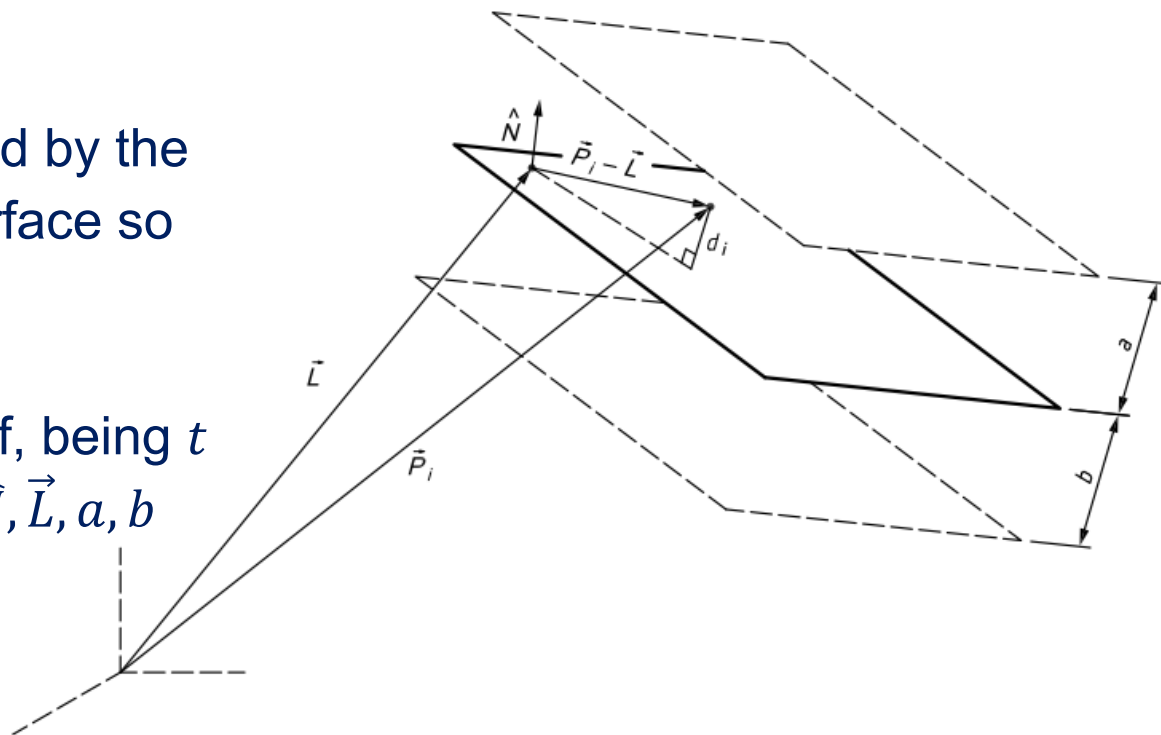


Definition of the tolerance zone (ISO 12781-1)

$$d_i = \hat{N} \cdot (\vec{P}_i - \vec{L})$$

The tolerance zone is defined by the set of $\vec{P}_i$ belonging to the surface so that $b \leq d_i \leq a$

The specification is verified if, being t the flatness tolerance, any $\hat{N}, \vec{L}, a, b$ exist so that $t \geq a - b$



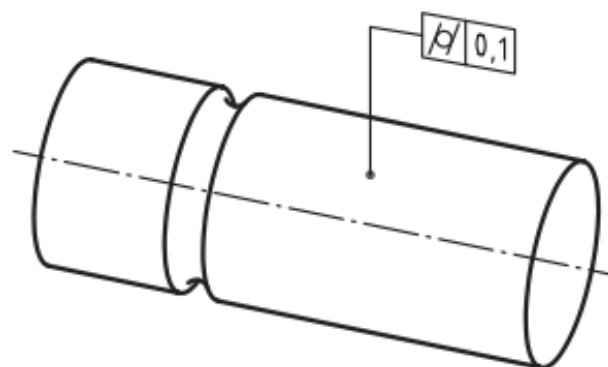
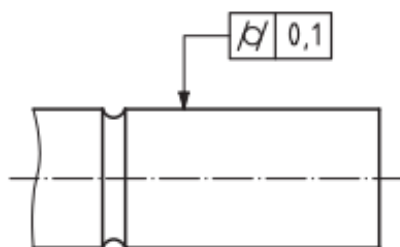


FORM TOLERANCES: CYLINDRICITY

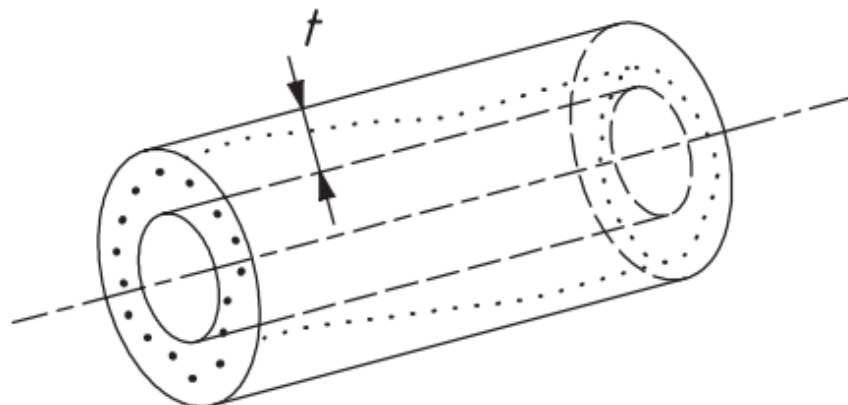
Applicability:

- Cylindrical surfaces

Indication



Interpretation



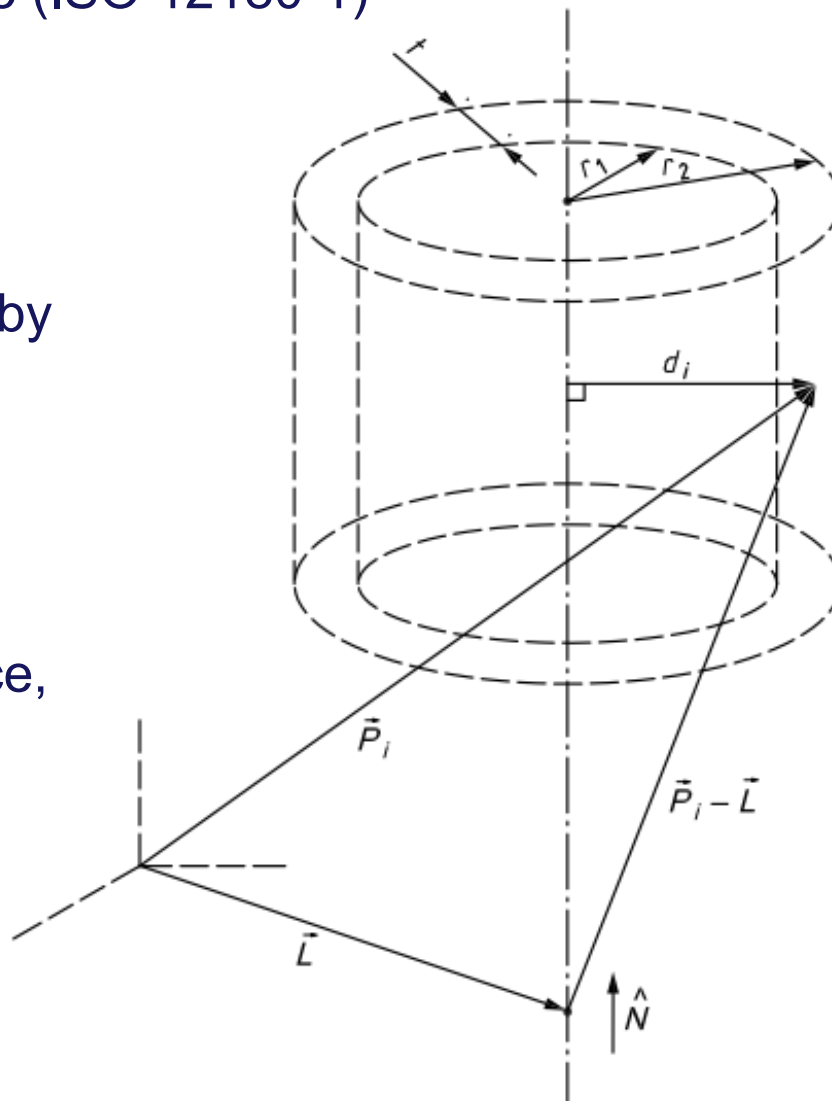
FORM TOLERANCES: CYLINDRICITY

Definition of the tolerance zone (ISO 12180-1)

$$d_i = |\hat{N} \times (\vec{P}_i - \vec{L})|$$

The tolerance zone is defined by the set of $\vec{P}_i$ belonging to the surface so that $r_1 \leq d_i \leq r_2$

The specification is verified if, being t the cylindricity tolerance, any $\hat{N}, \vec{L}, r_1, r_2$ exist so that $t \geq r_2 - r_1$

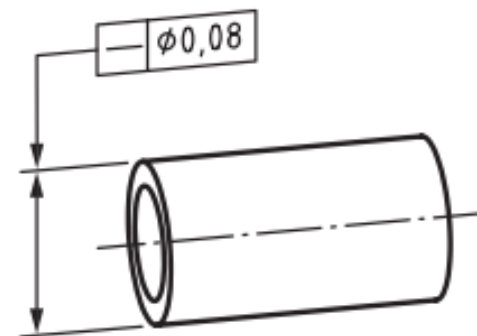
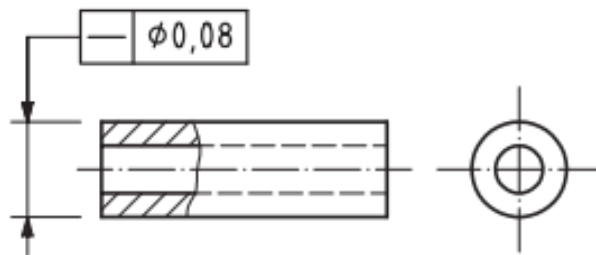


FORM TOLERANCES: STRAIGHTNESS

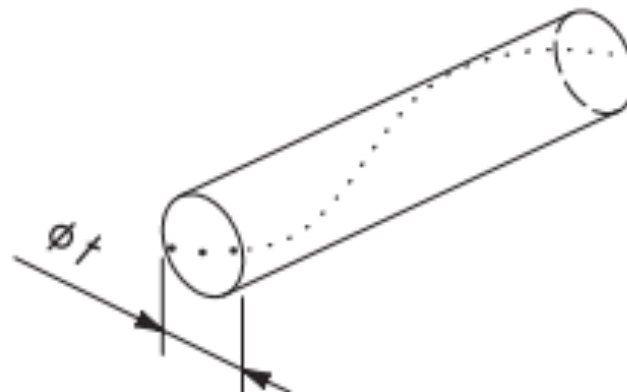
Applicability:

- Integral features (surface of planes, cylinders, cones,...)
- **Derived features** (axes of cylinders, median planes,...)

Indication:



Interpretation:



FORM TOLERANCES: STRAIGHTNESS

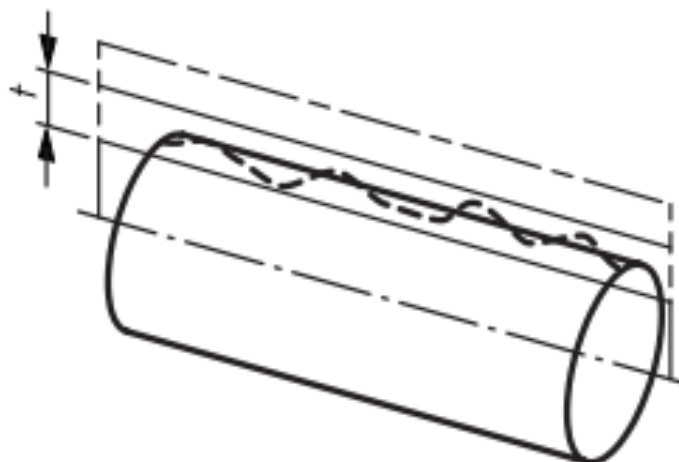
Applicability:

- **Integral features** (surface of planes, cylinders, cones,...)
- Derived features (axes of cylinders, median planes,...)

Indication:



Interpretation:

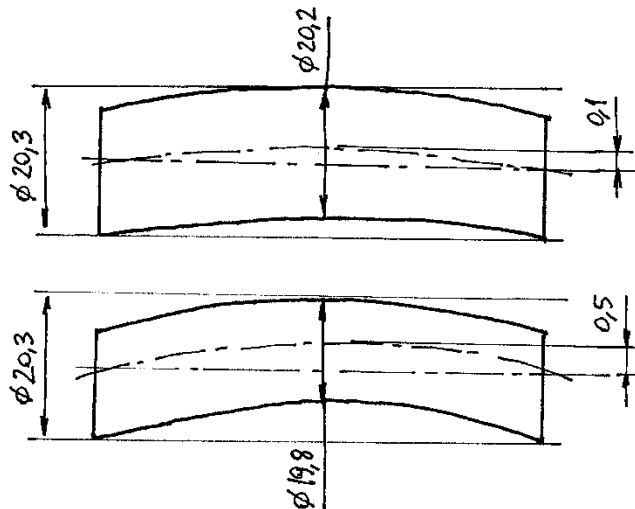
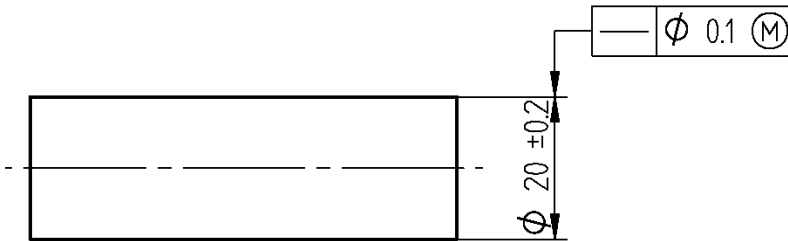




FORM TOLERANCES: STRAIGHTNESS - MMC

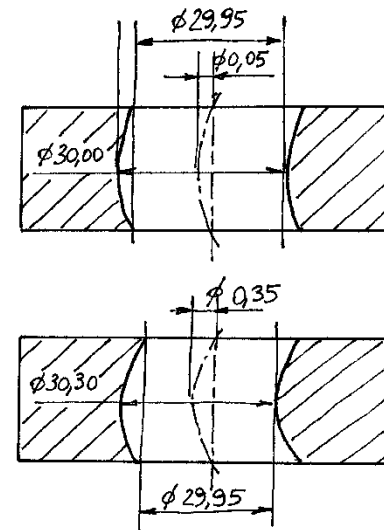
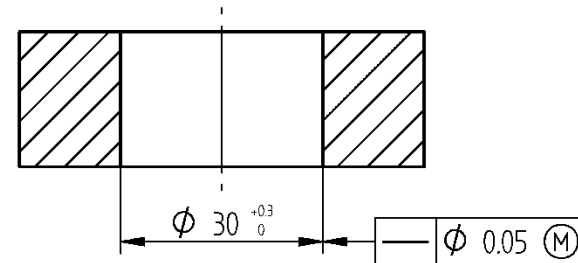
MMC: Maximum material condition

Application to derived features with bonus



bonus = MMC – measured size

Virtual condition: MMC + tolerance

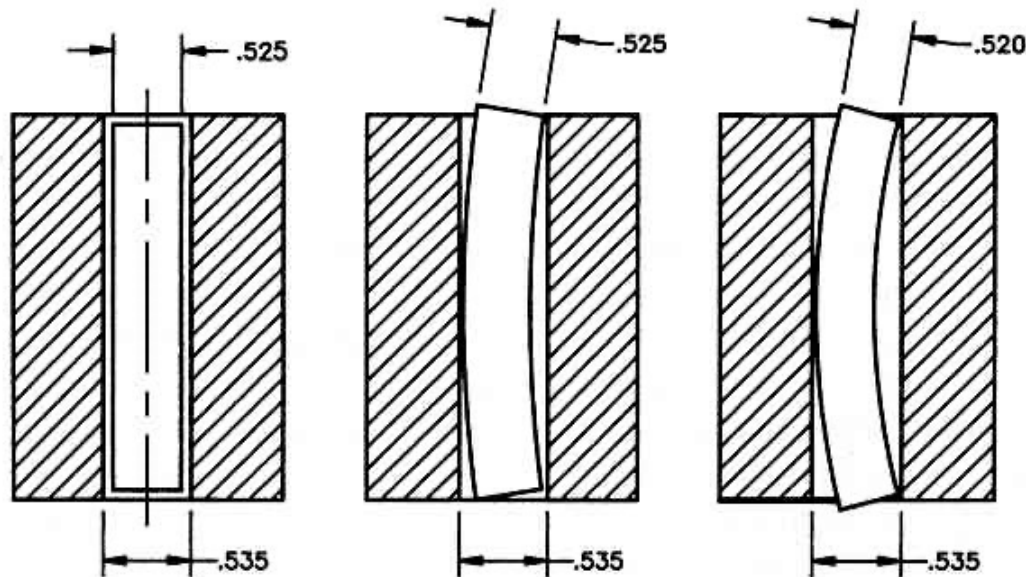
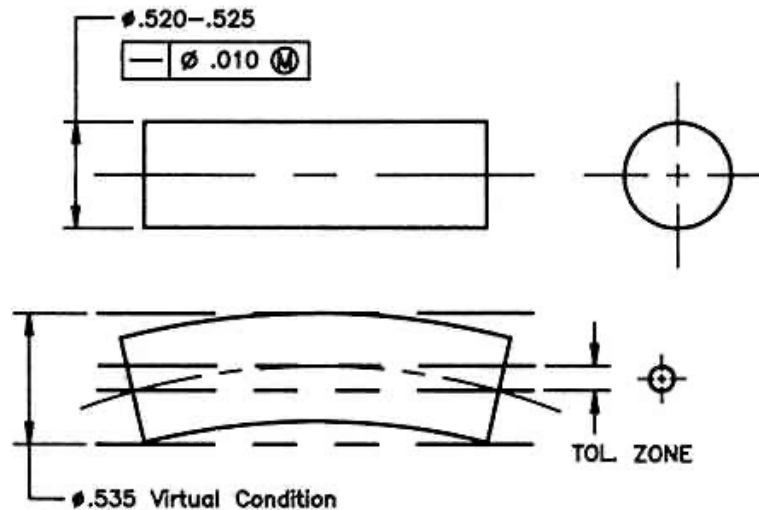


bonus = measured size - MMC

Virtual condition: MMC - tolerance



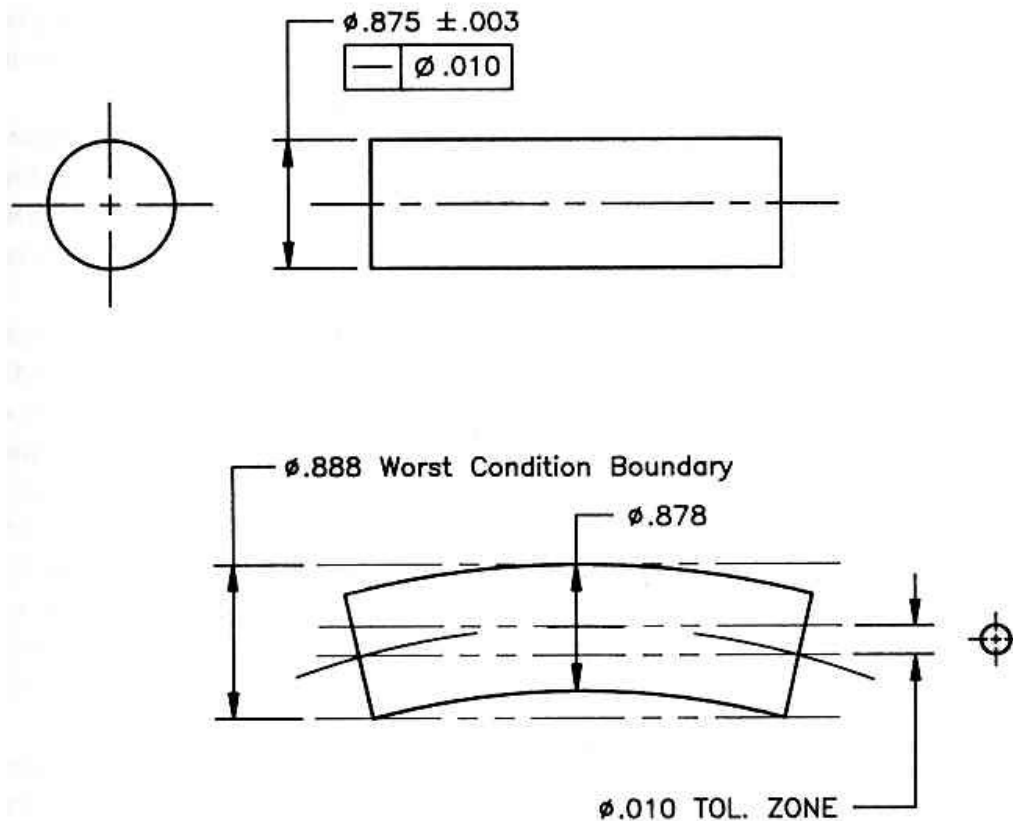
FORM TOLERANCES: STRAIGHTNESS - MMC



Diameter	Geo. Tol.
.525	.010
.524	.011
.523	.012
.522	.013
.521	.014
.520	.015



FORM TOLERANCES: STRAIGHTNESS - MMC



Diameter	Geo. Tol.
.872	.010
.873	.010
.874	.010
.875	.010
.876	.010
.877	.010
.878	.010

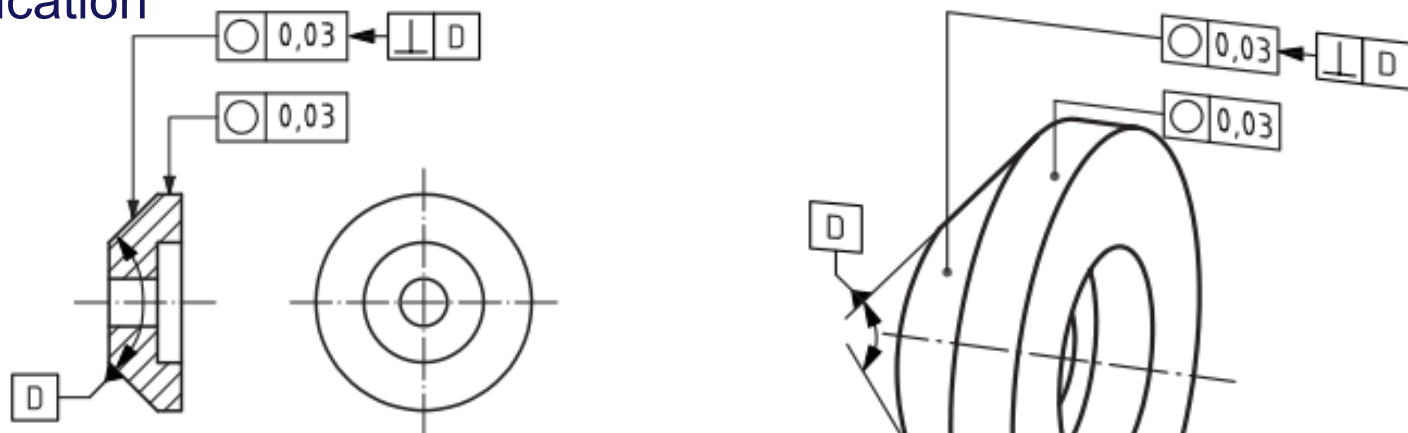


FORM TOLERANCES: ROUNDNESS

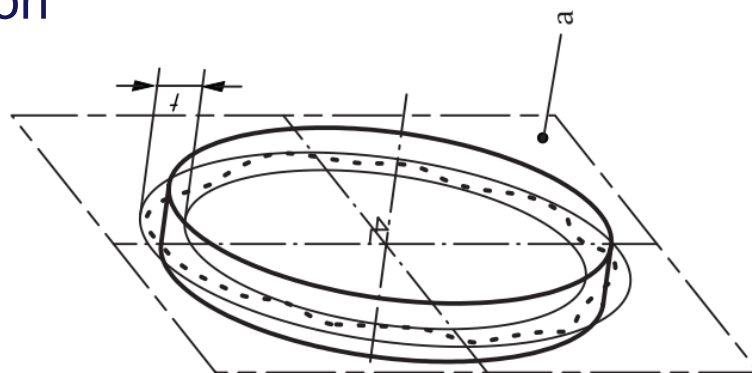
Applicability:

- Circular surfaces

Indication



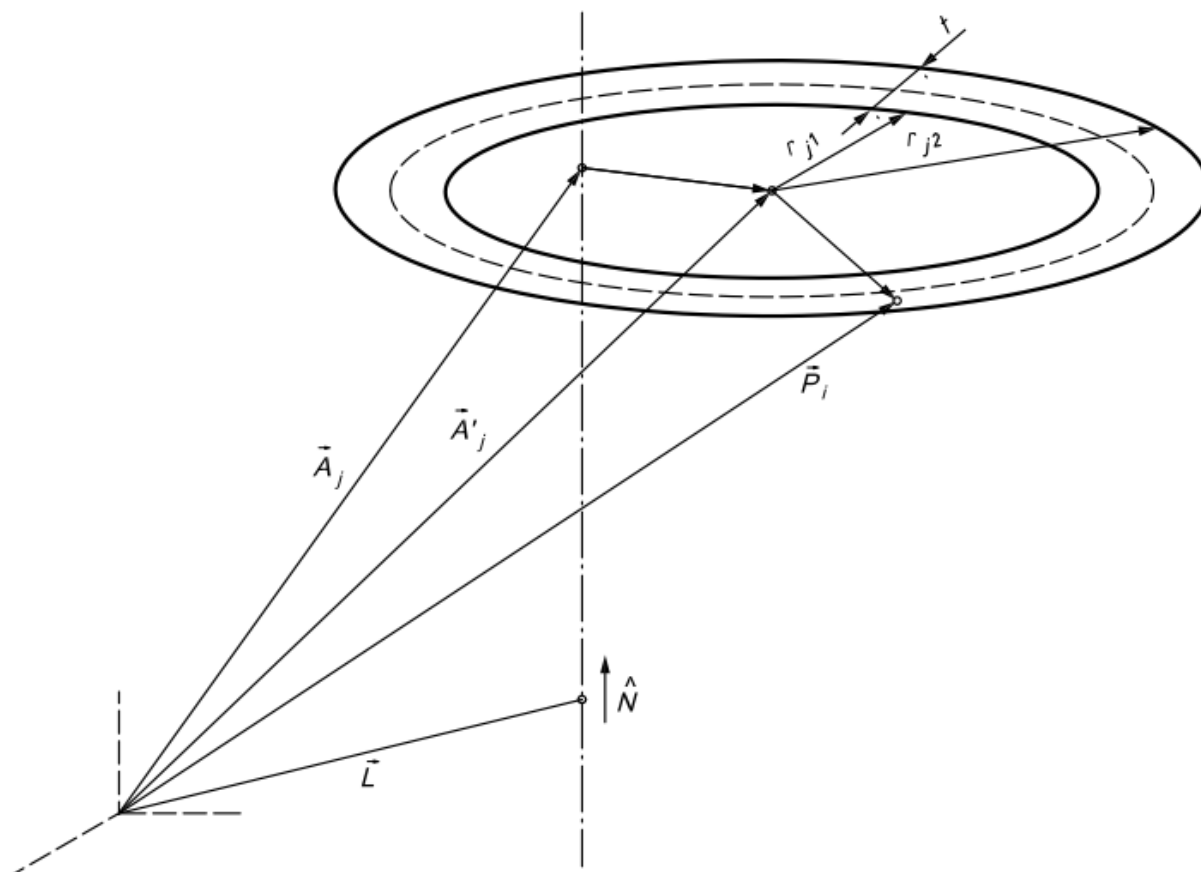
Interpretation





Definition of the tolerance zone (ISO 12181-1)

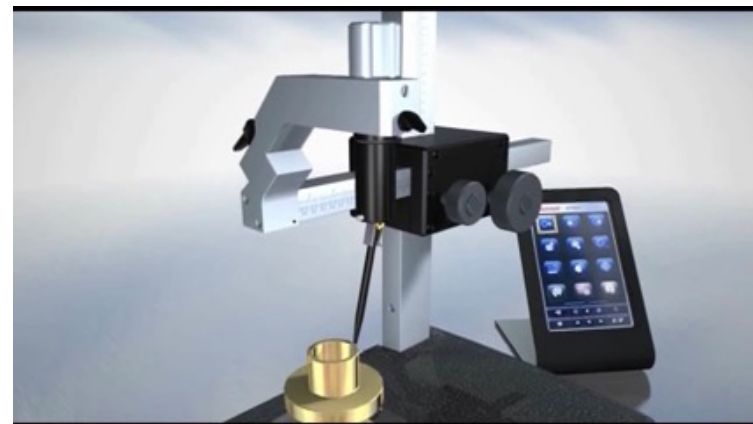
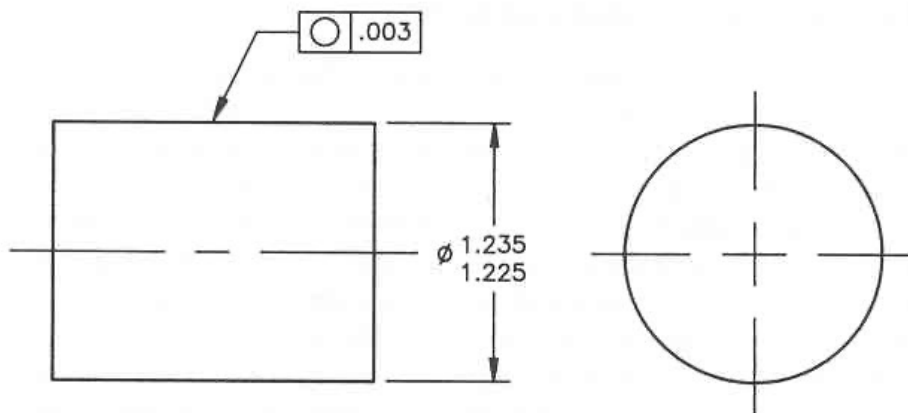
$$\begin{aligned}\vec{A}_j &= \vec{L} + c_j \hat{N} \\ (\vec{A}'_j - \vec{A}_j) \cdot \hat{N} &= 0 \\ (\vec{P}_i - \vec{A}'_j) \cdot \hat{N} &= 0 \\ r_{j1} &\leq |\vec{P}_i - \vec{A}'_j| \leq r_{j2} \\ r_{j2} - r_{j1} &\leq t\end{aligned}$$



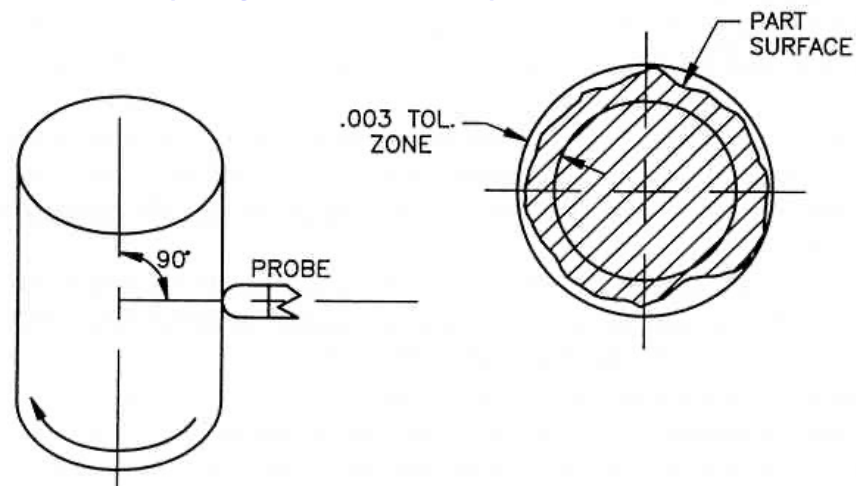
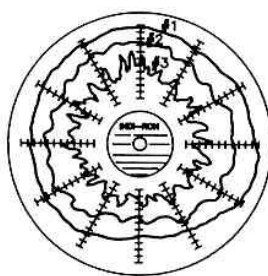
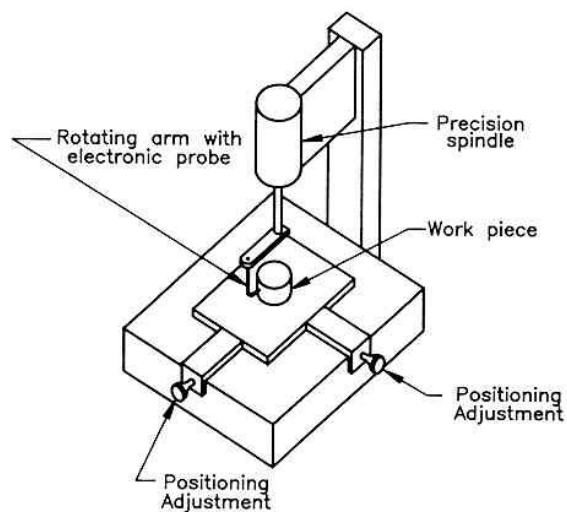


FORM TOLERANCES: ROUNDNESS

30

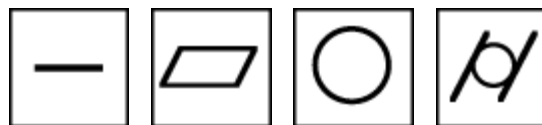


<https://youtu.be/rAKbqTloS4k>

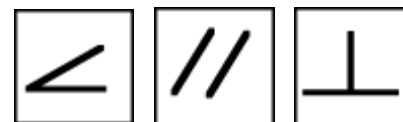




Form



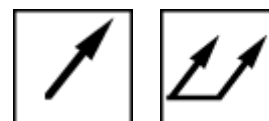
Orientation



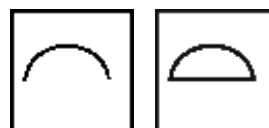
Location



Run-out



Profile

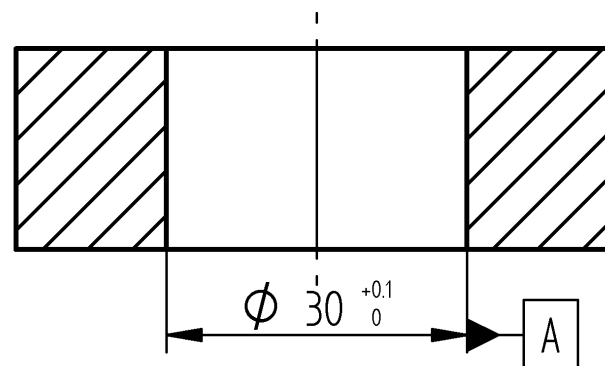
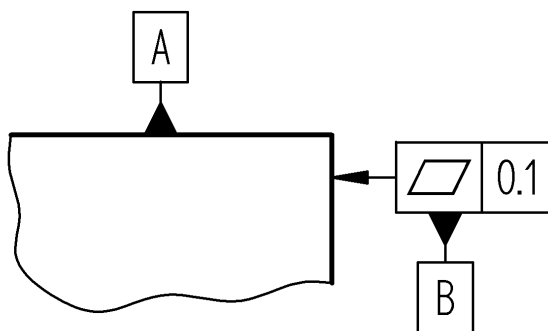




GEOMETRIC TOLERANCES - DATUMS

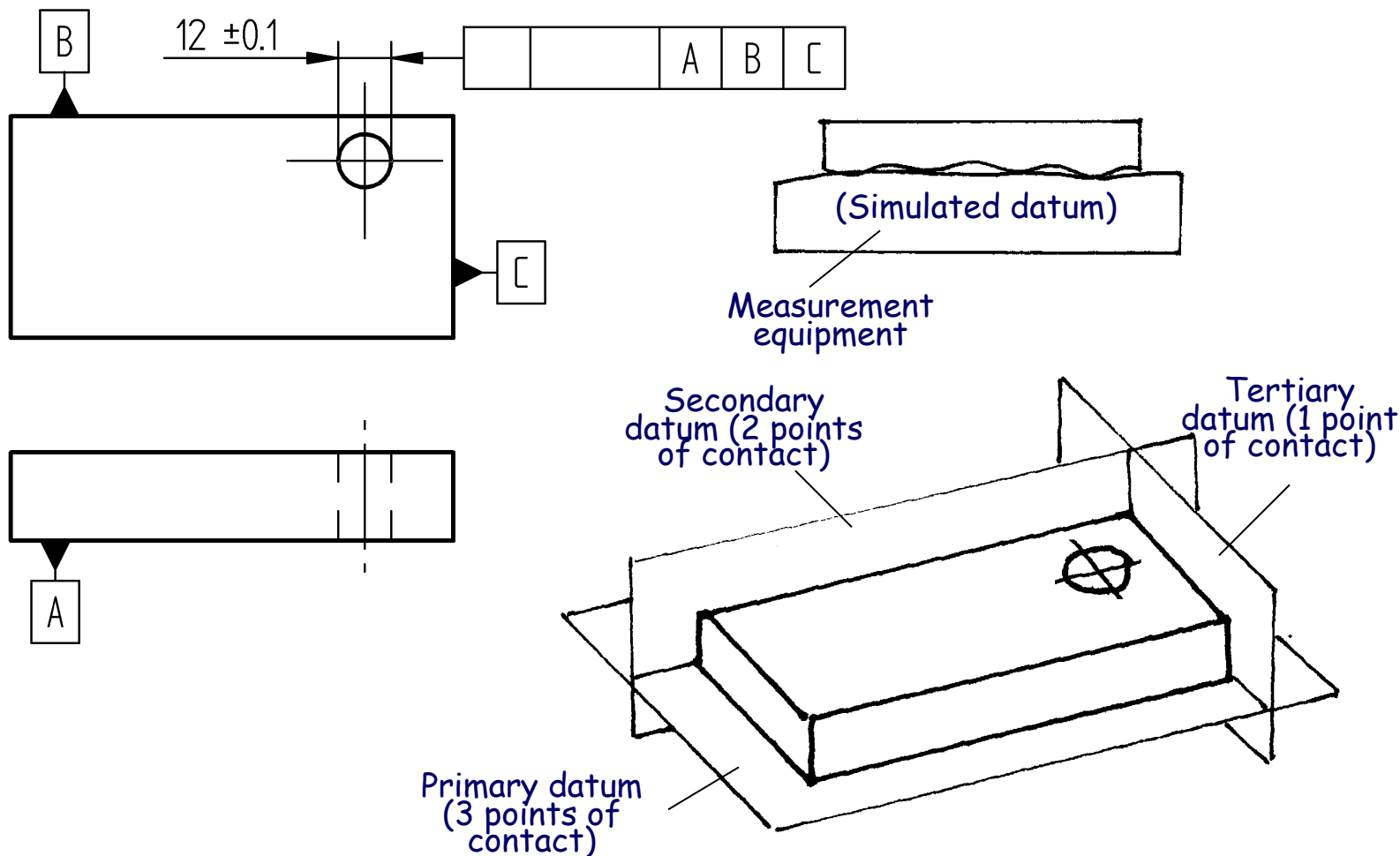
Definitions (ISO 5459):

Datum	one or more situation features of one or more features associated with one or more real integral features selected to define the location or orientation, or both, of a tolerance zone or an ideal feature representing for instance a virtual condition
Datum Feature	real (non-ideal) integral feature used for establishing a datum
Simulated Datum	surfaces that reproduce the references in the measurement phase



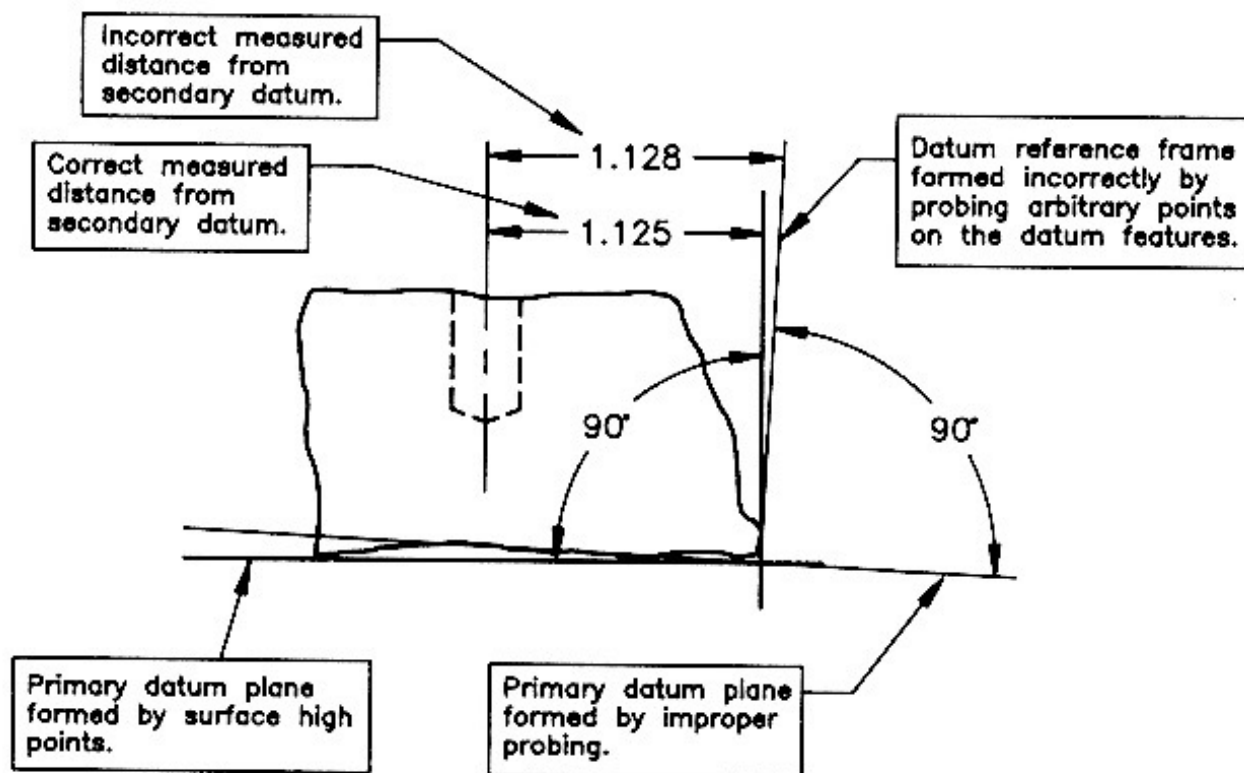


Datum systems



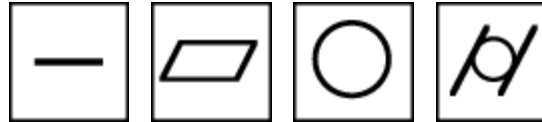


GEOMETRIC TOLERANCES - DATUMS

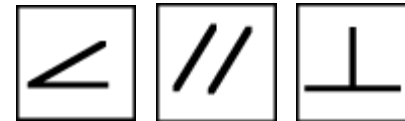




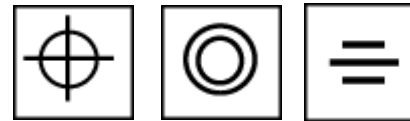
Form



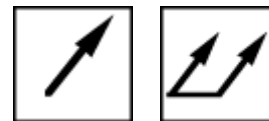
Orientation



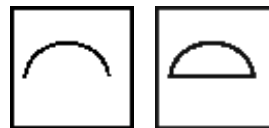
Location



Run-out



Profile



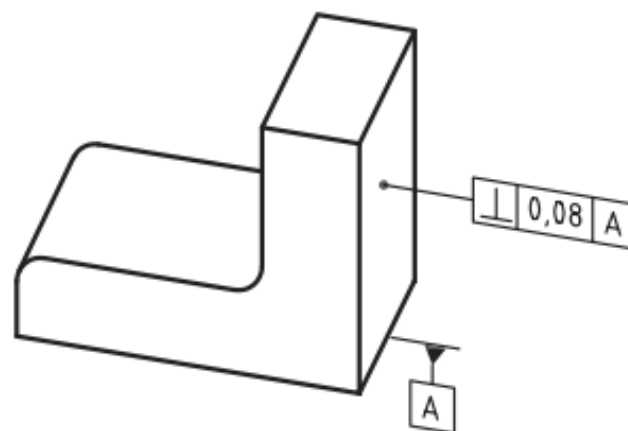
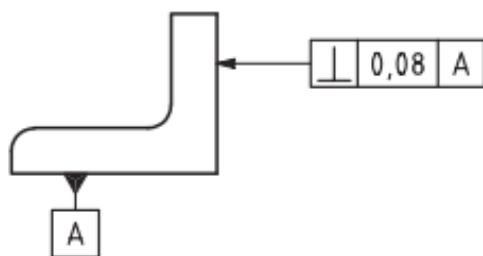


ORIENTATION TOLERANCES: PERPENDICULARITY

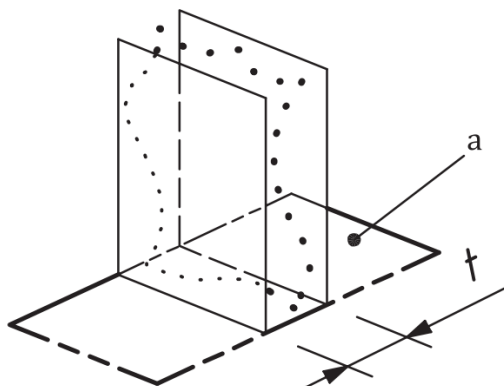
Applicability:

- **Integral features** (Flat surfaces)
- Derived features (axes, median planes)

Indication:



Interpretation:



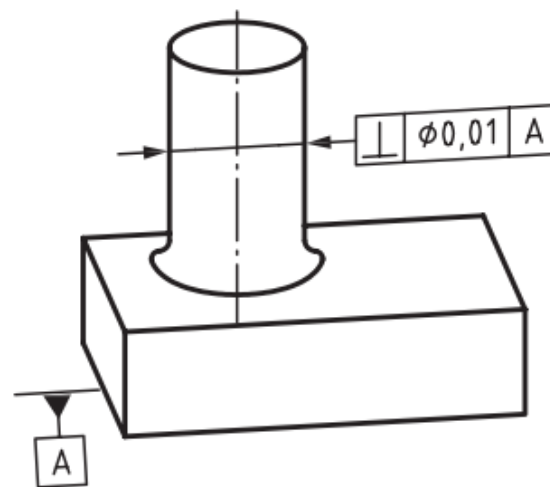
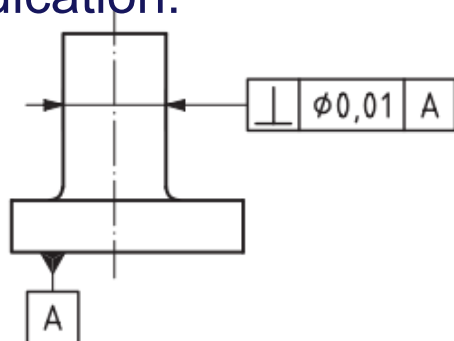


ORIENTATION TOLERANCES: PERPENDICULARITY

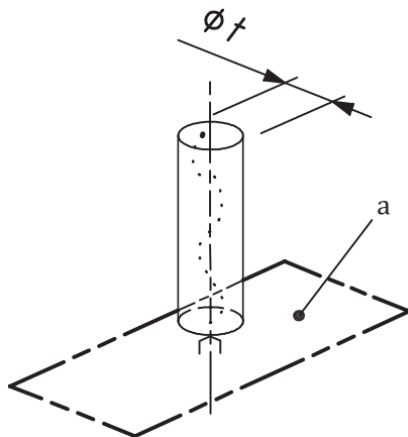
Applicability:

- Integral features (Flat surfaces)
- **Derived features** (axes, median planes)

Indication:



Interpretation:



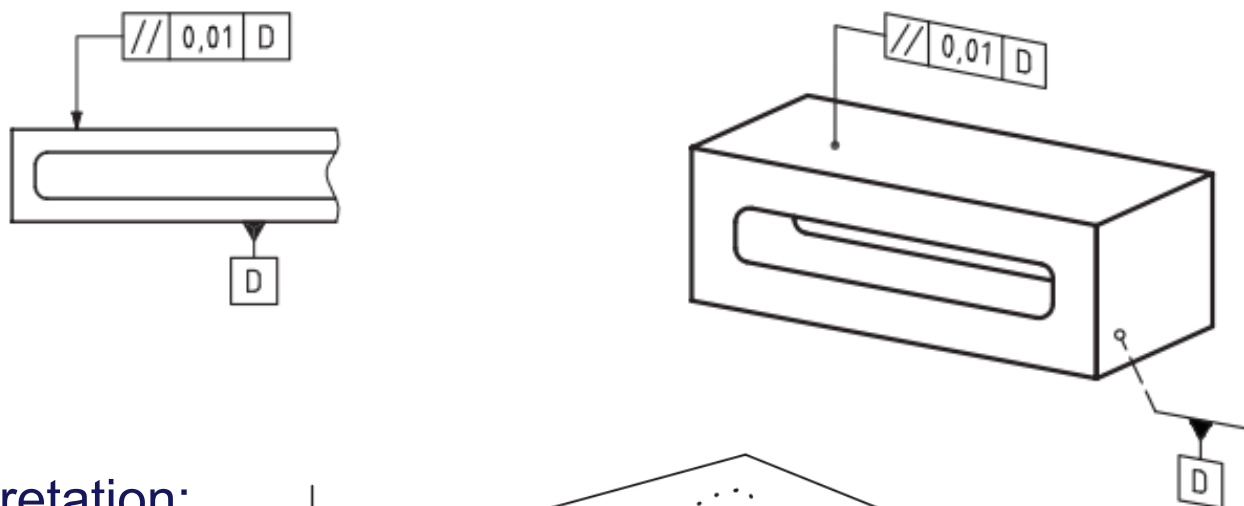


ORIENTATION TOLERANCES: PARALLELISM

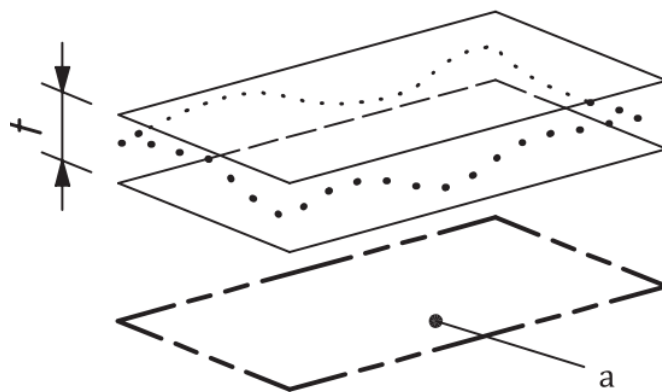
Applicability:

- **Integral features** (Flat surfaces)
- Derived features (axes, median planes)

Indication:



Interpretation:



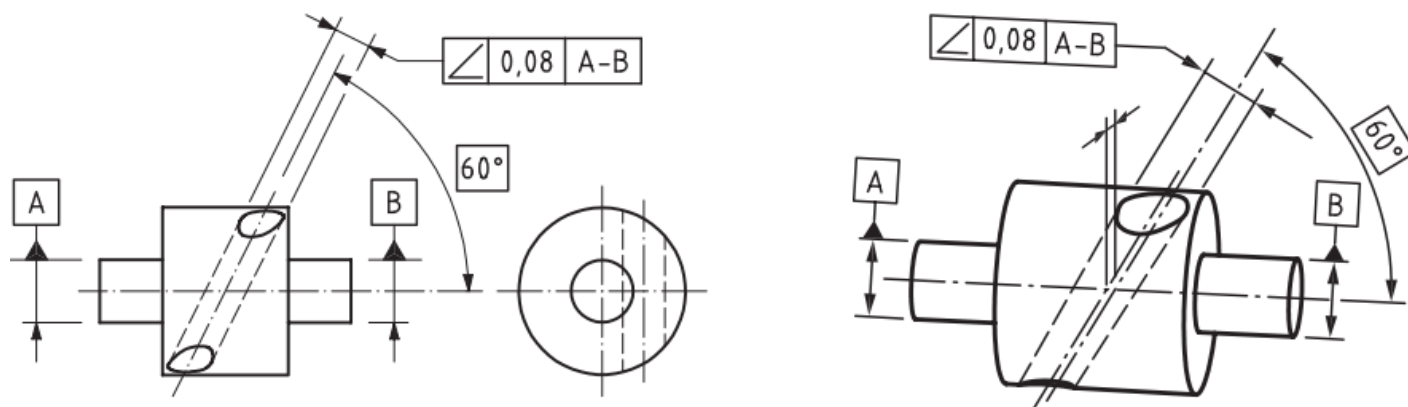


ORIENTATION TOLERANCES: ANGULARITY

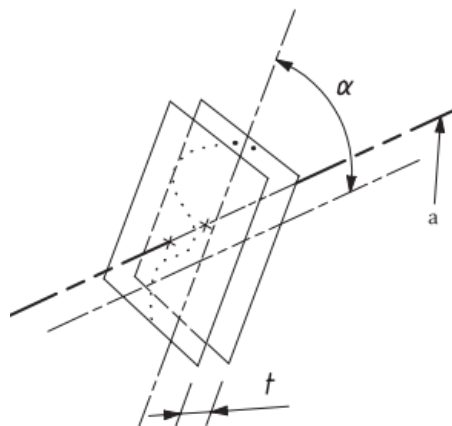
Applicability:

- Integral features (Flat surfaces)
- **Derived features** (axes, median planes)

Indication:

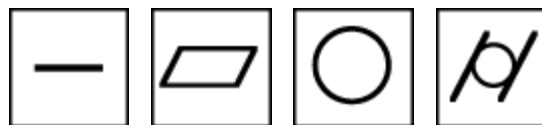


Interpretation:

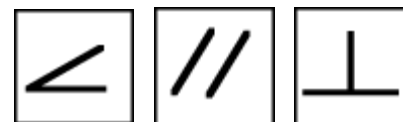




Form



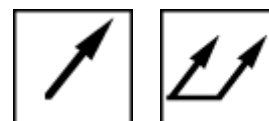
Orientation



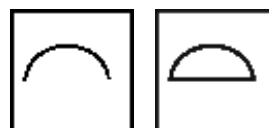
Location



Run-out



Profile

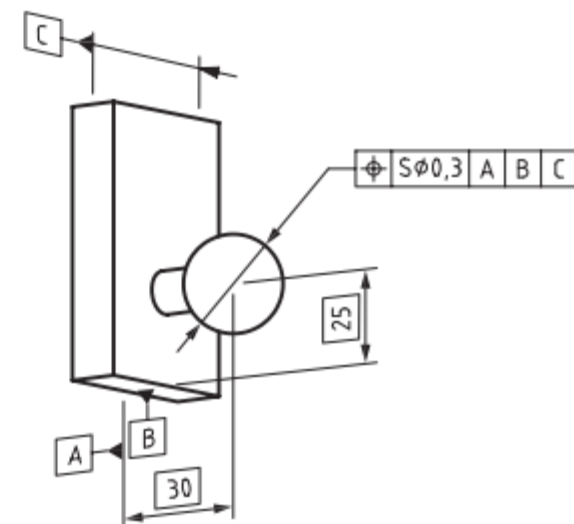
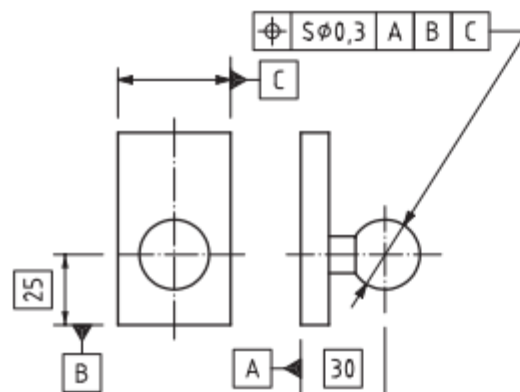




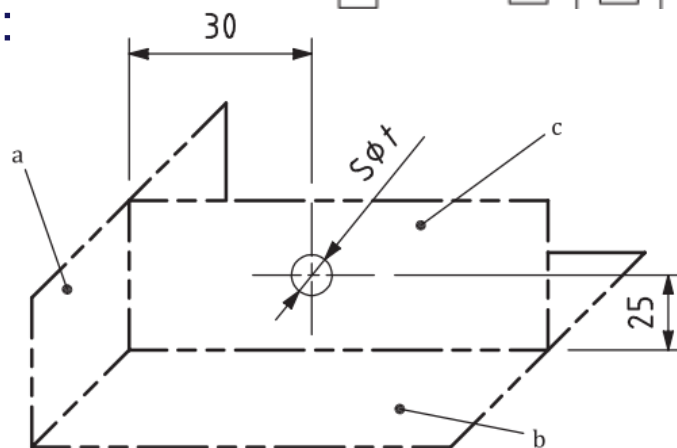
Applicability:

- Derived features from features of size (axes, median planes, sphere center)
- In the ISO standards only: line, planes

Indication:



Interpretation:



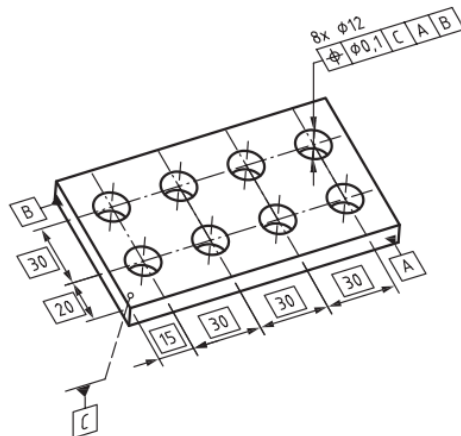
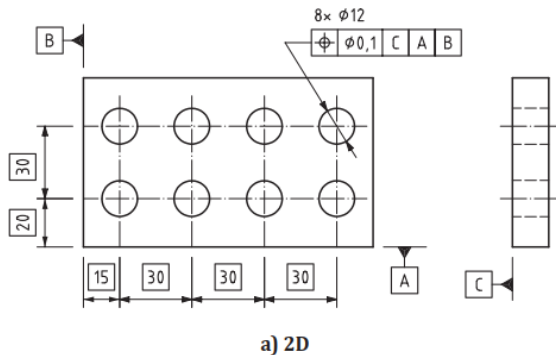


LOCATION TOLERANCES: POSITION

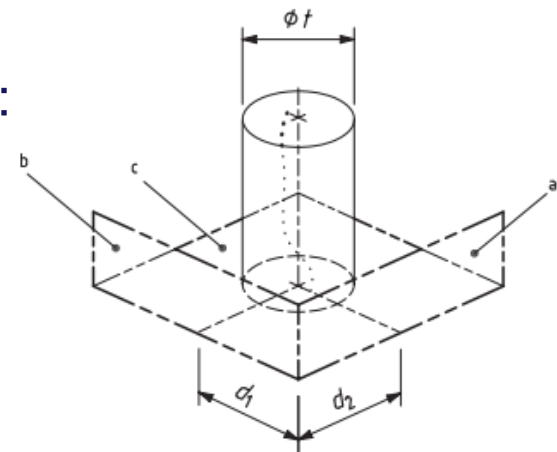
Applicability:

- Derived features from features of size (axes, median planes, sphere center)
- In the ISO standards only: line, planes

Indication:

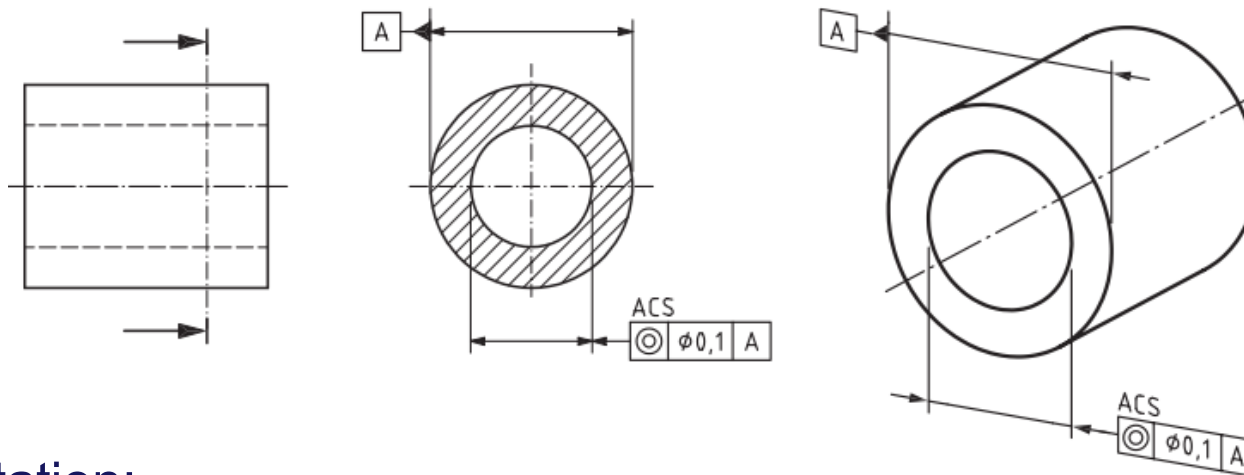


Interpretation:

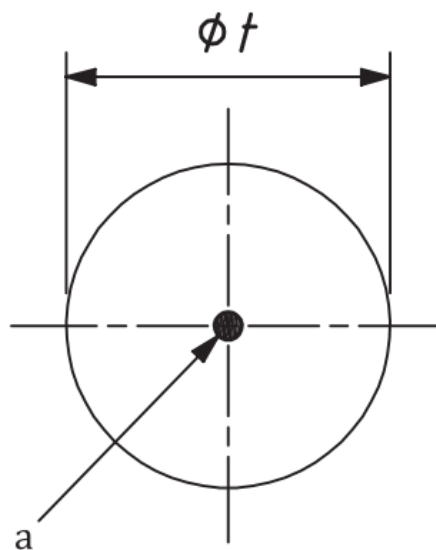




Indication:



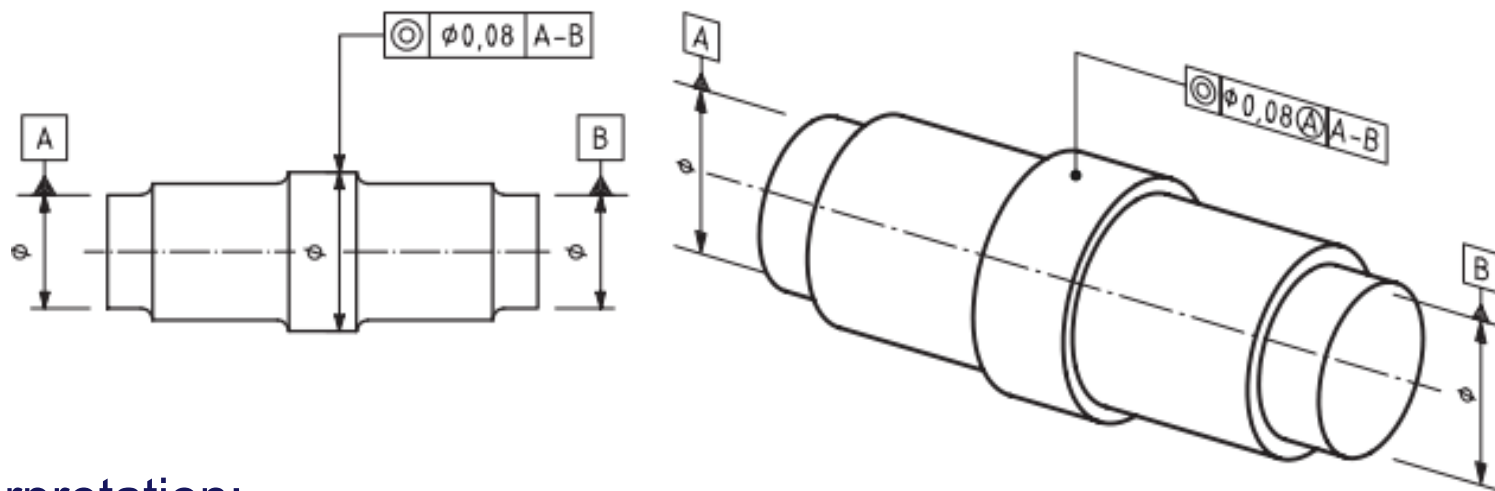
Interpretation:



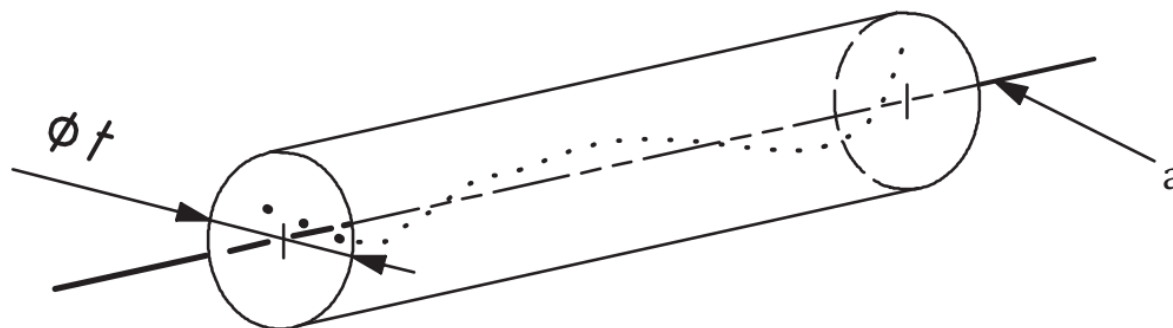


LOCATION TOLERANCES: COAXIALITY

Indication:



Interpretation:



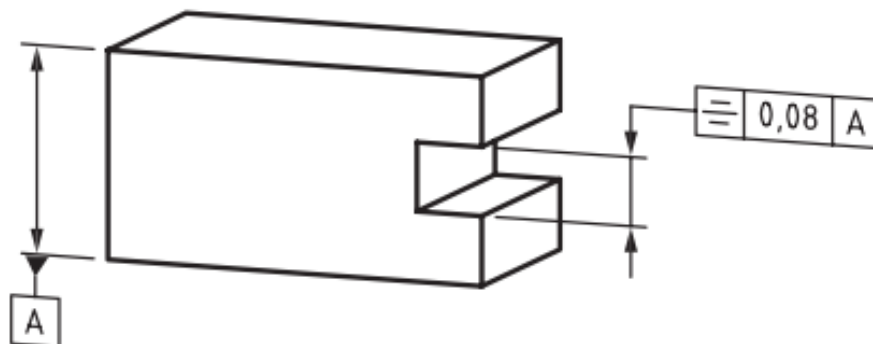
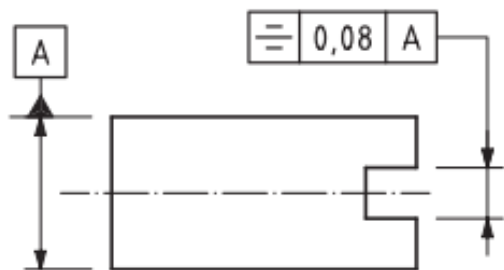


LOCATION TOLERANCES: SYMMETRY

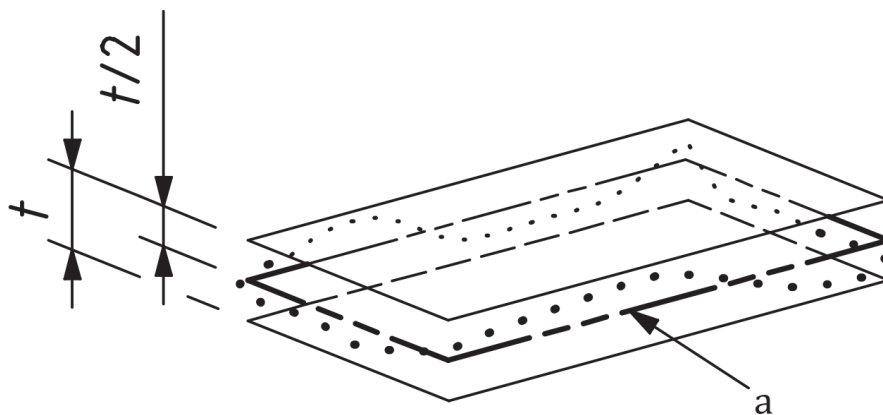
Applicability

- Features of size (axes, median planes)

Indication:

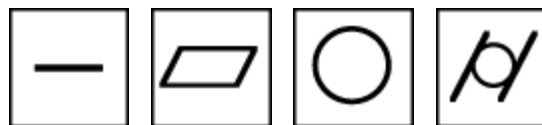


Interpretation:

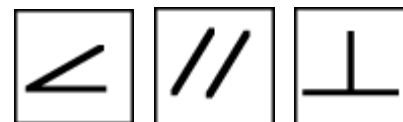




Form



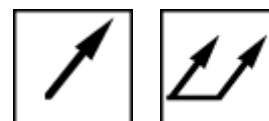
Orientation



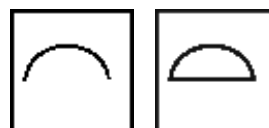
Location



Run-out



Profile



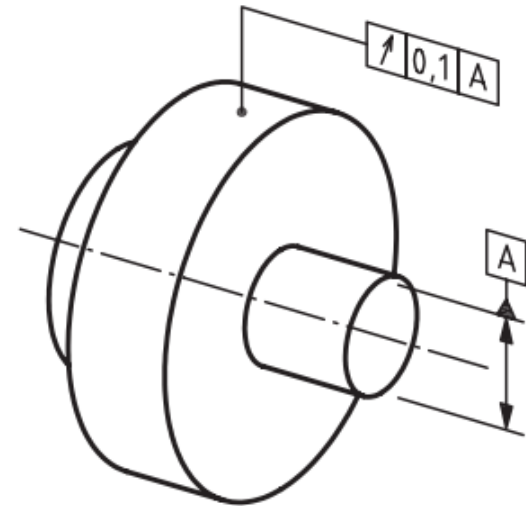
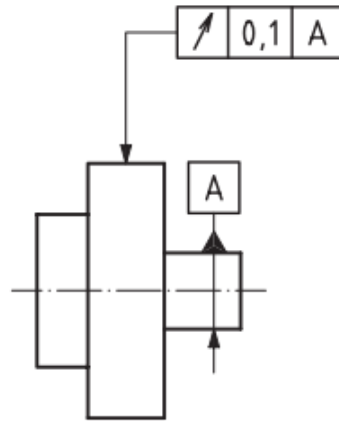


RUN-OUT TOLERANCES: CIRCULAR RUN-OUT

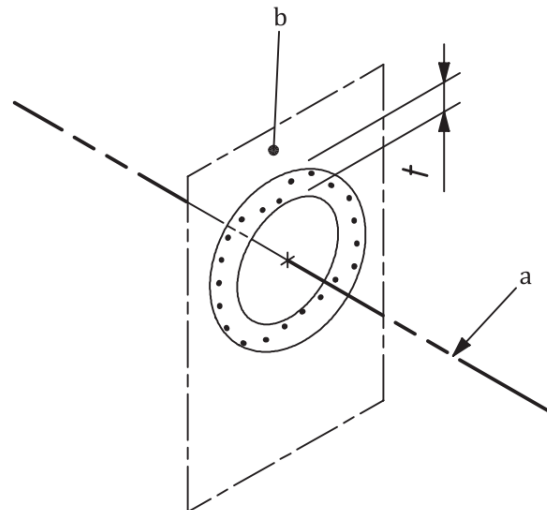
Applicability:

- **Circular** and flat surfaces

Indication:



Interpretation:



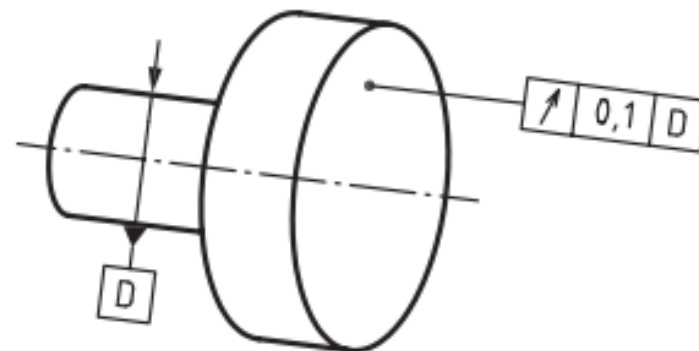
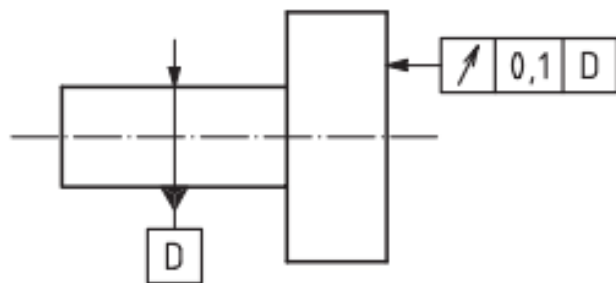


RUN-OUT TOLERANCES: CIRCULAR RUN-OUT

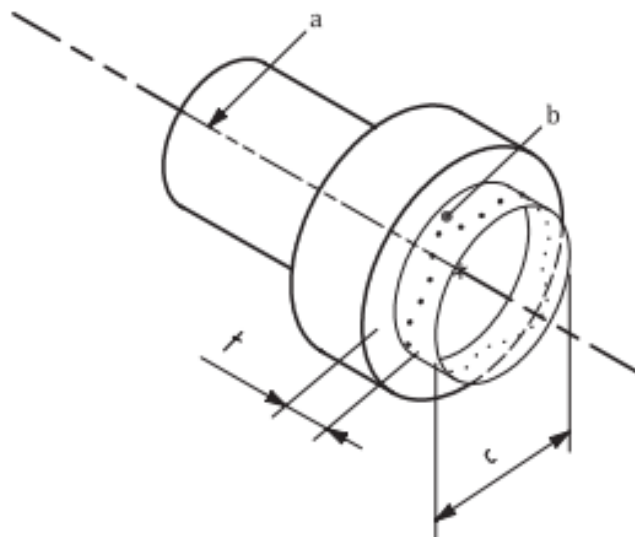
Applicability:

- Circular and **flat** surfaces

Indication:



Interpretation:



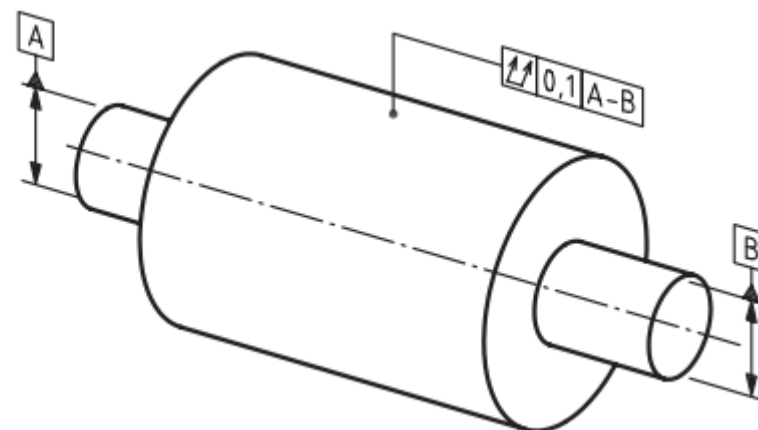
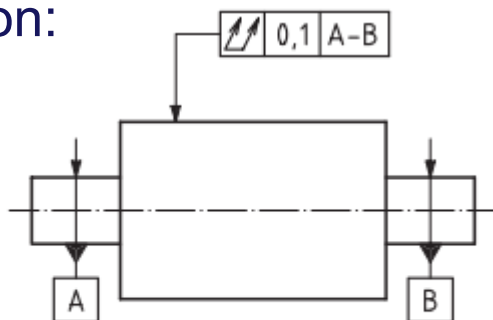


RUN-OUT TOLERANCES: TOTAL RUN-OUT

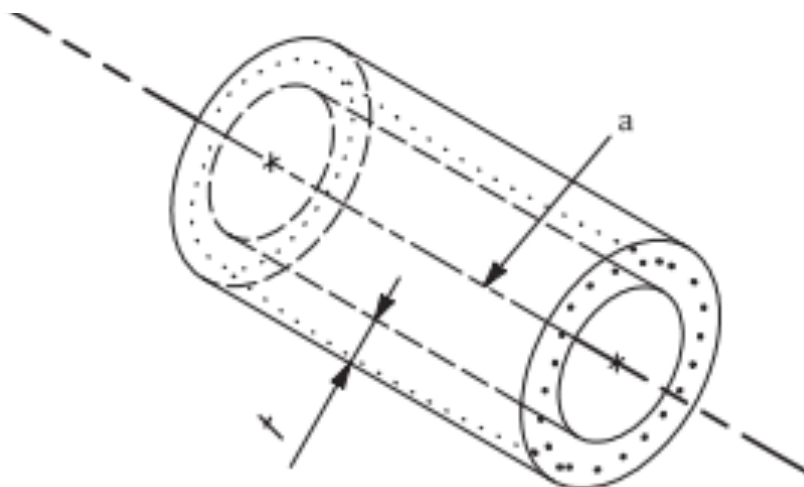
Applicability:

- **Circular** and flat surfaces

Indication:



Interpretation:



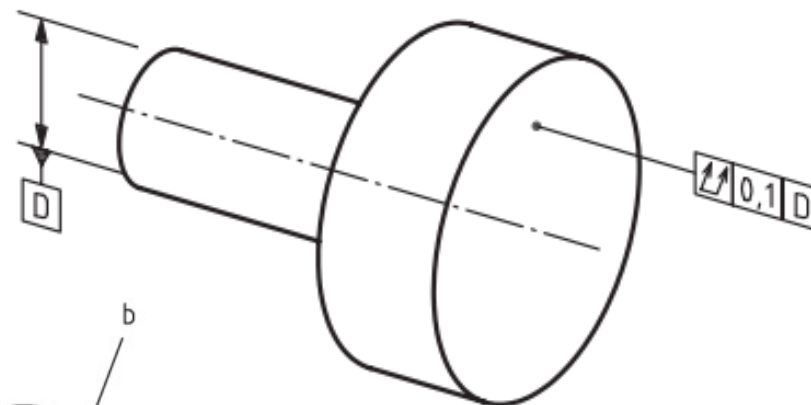
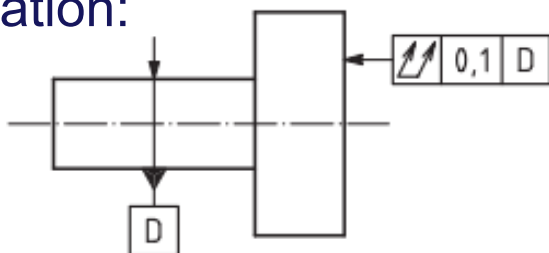


RUN-OUT TOLERANCES: TOTAL RUN-OUT

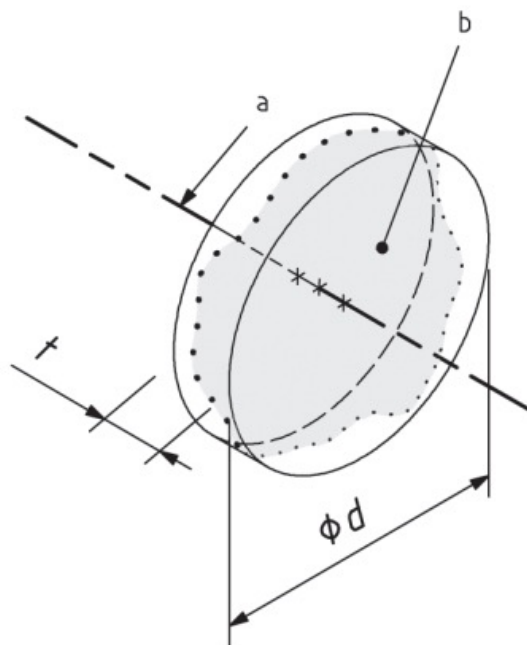
Applicability:

- Circular and **flat** surfaces

Indication:

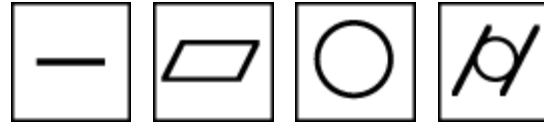


Interpretation:

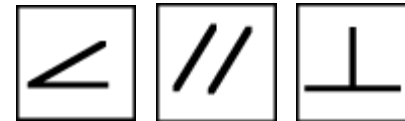




Form



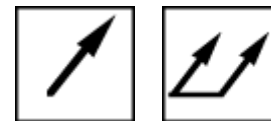
Orientation



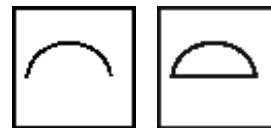
Location



Run-out



Profile



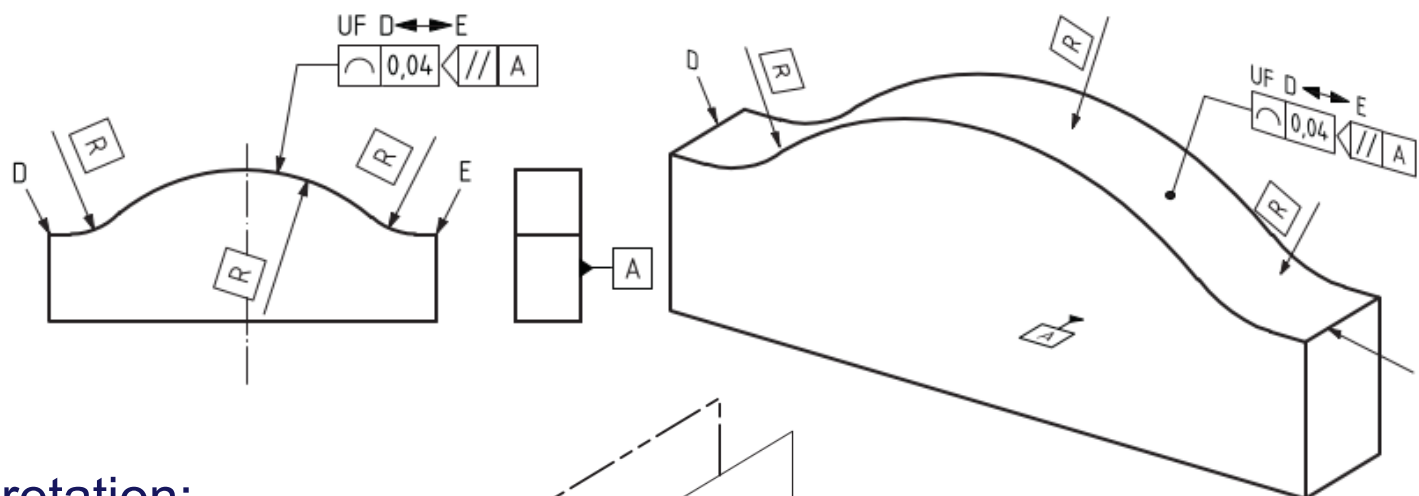


PROFILE TOLERANCE – LINES WITHOUT DATUM

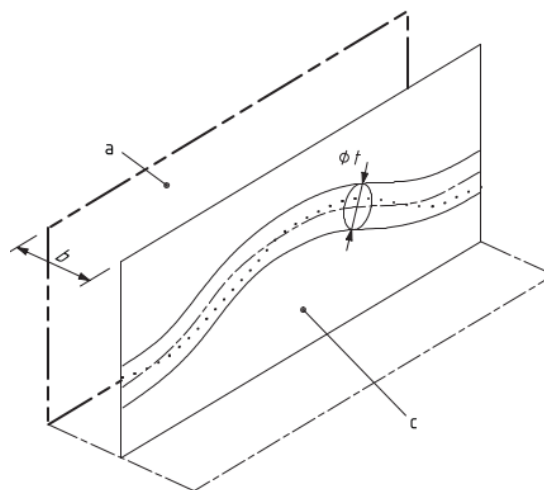
Applicability:

- Surfaces

Indication:



Interpretation:



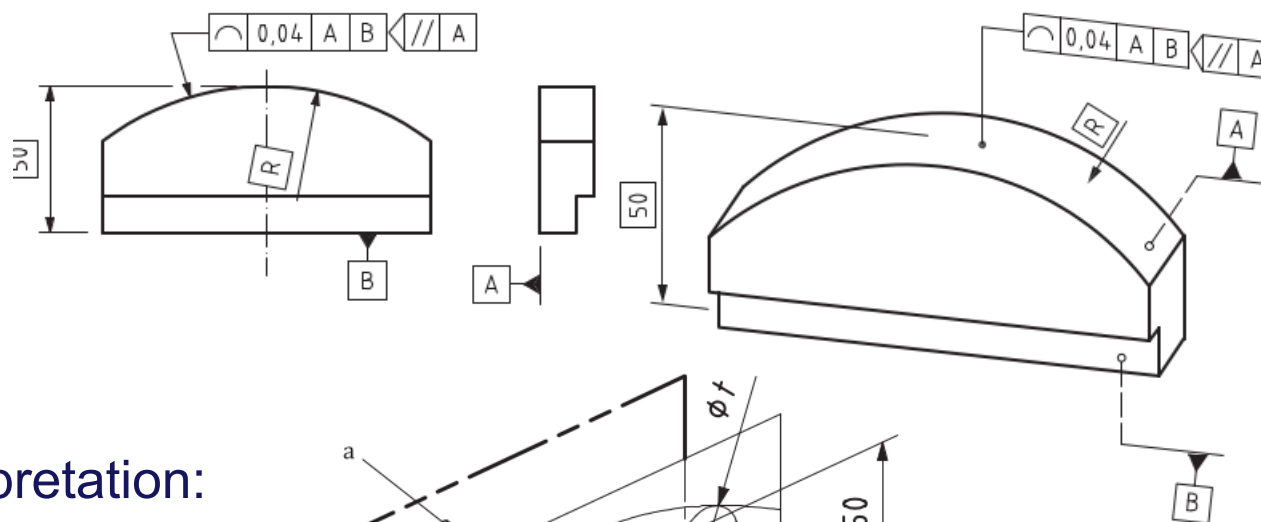


PROFILE TOLERANCE – LINES WITH DATUM

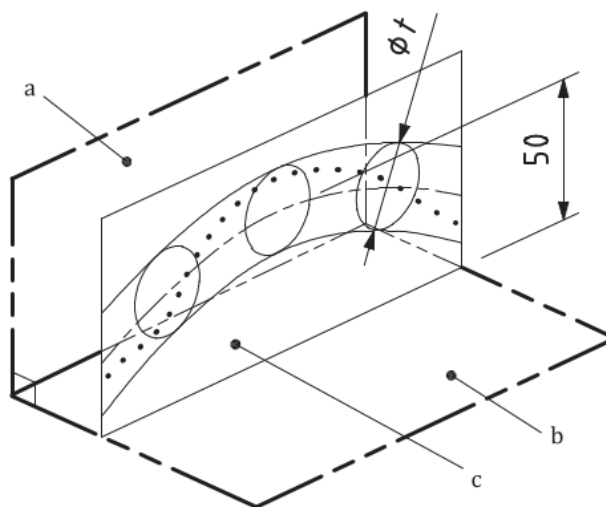
Applicability:

- Surfaces

Indication:



Interpretation:



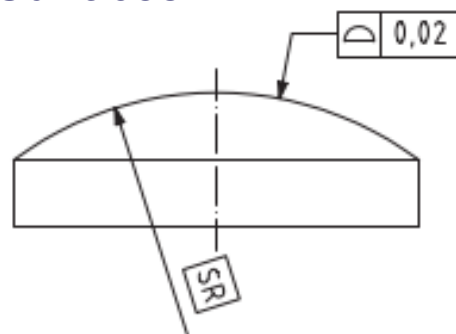


PROFILE TOLERANCE – SURFACES WITHOUT DATUM

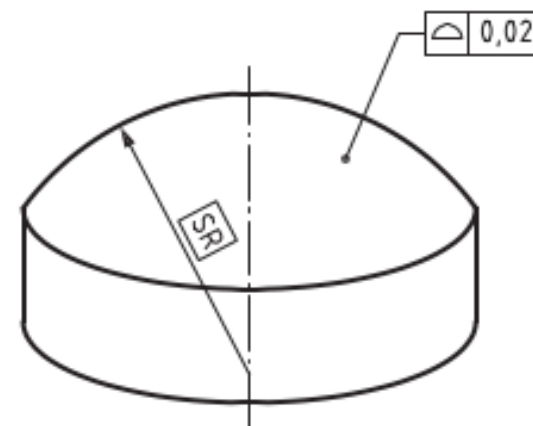
Applicability:

- Surfaces

Indication:

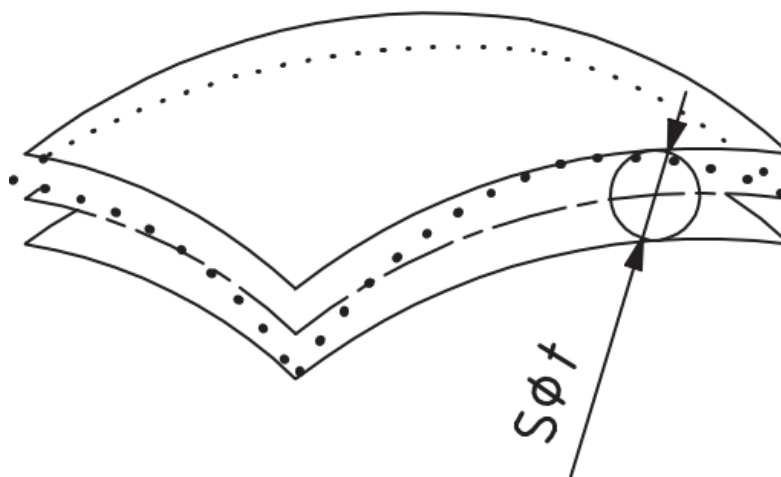


a) 2D



b) 3D

Interpretation:



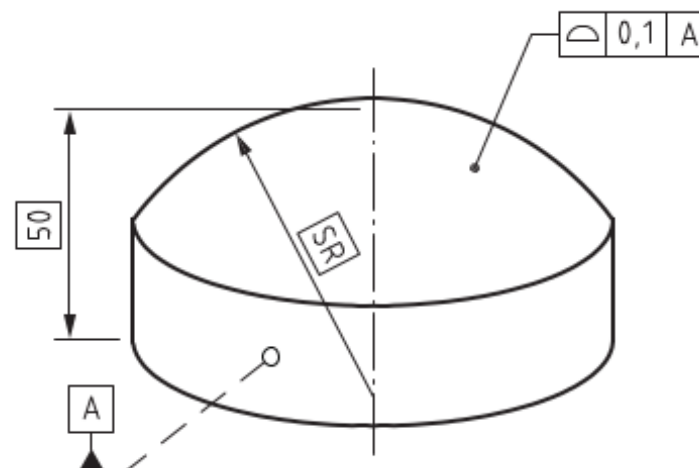
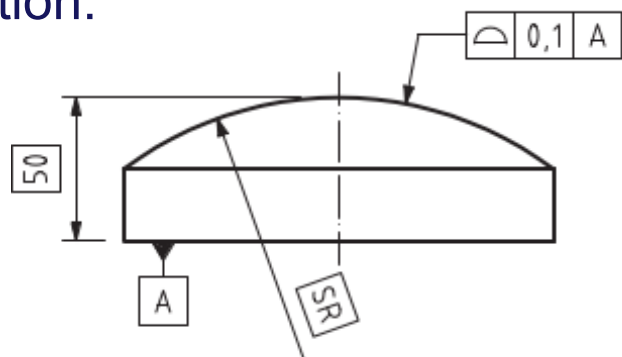


PROFILE TOLERANCE – SURFACES WITH DATUM

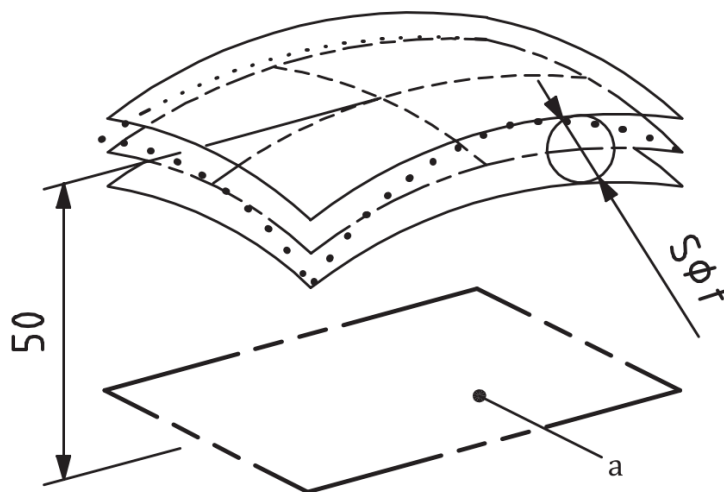
Applicability:

- Surfaces

Indication:



Interpretation:





STANDARD FRAMEWORK: ISO vs. ASME

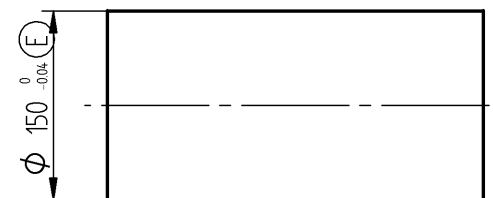
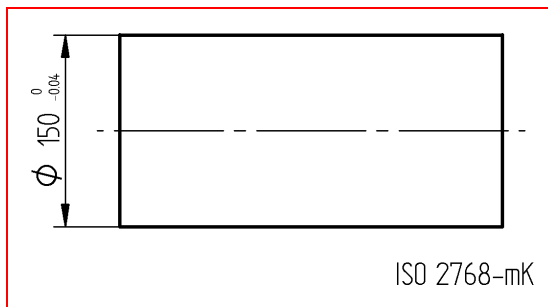
56



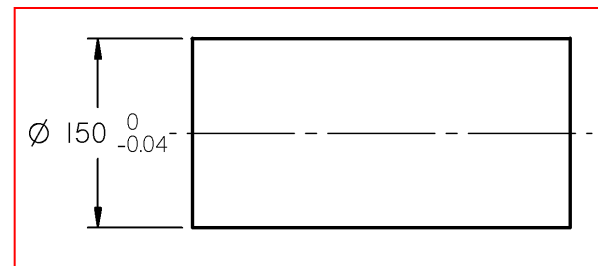
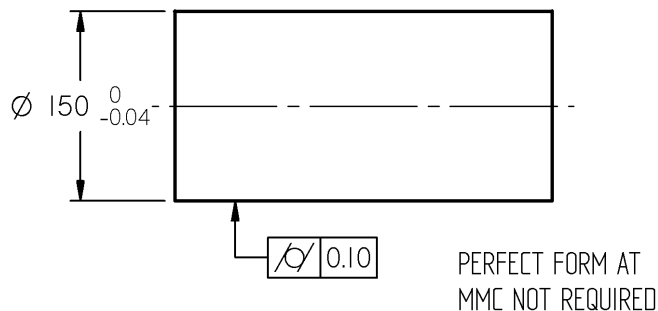
Independence
(size does not control form)

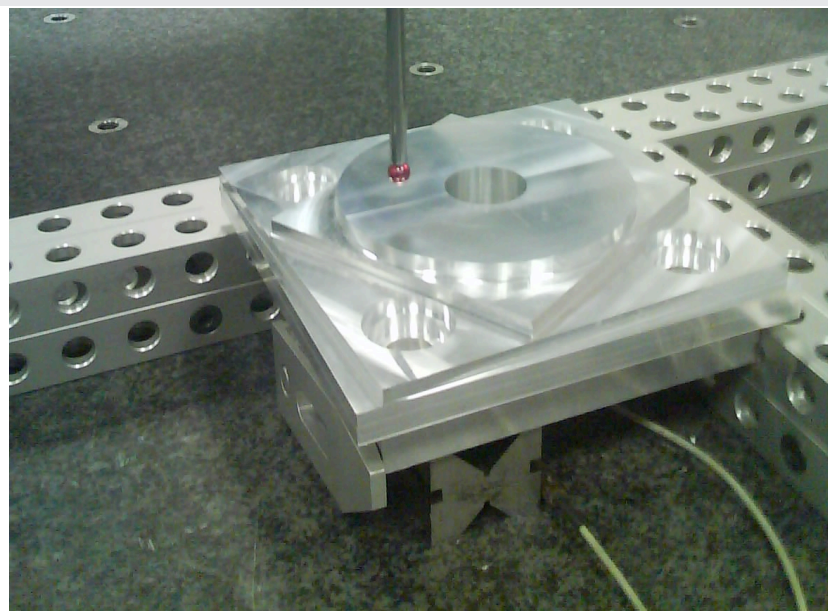
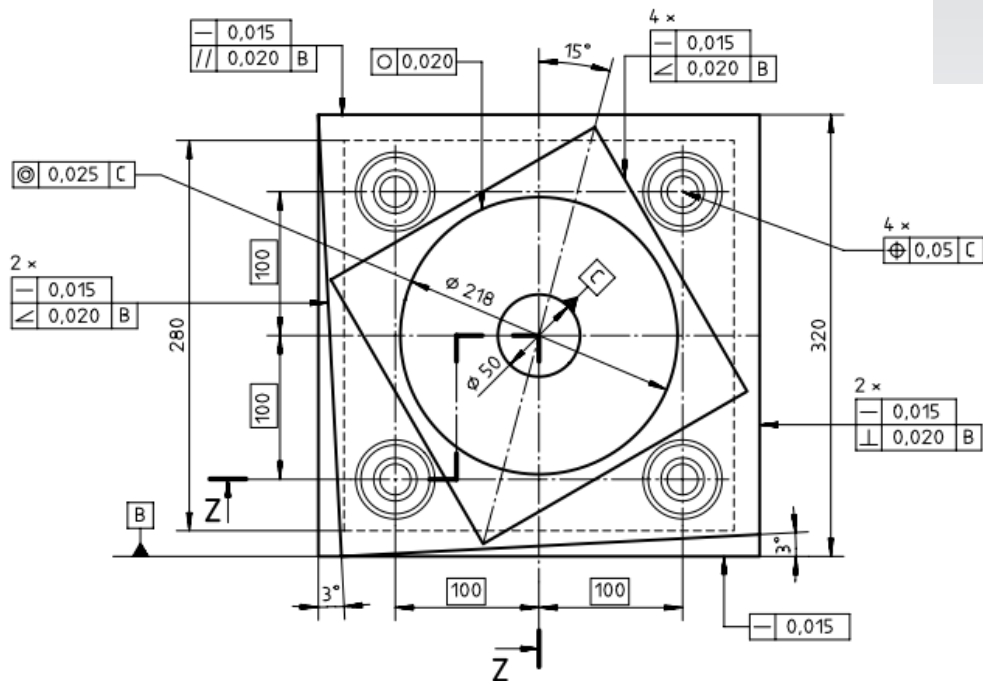
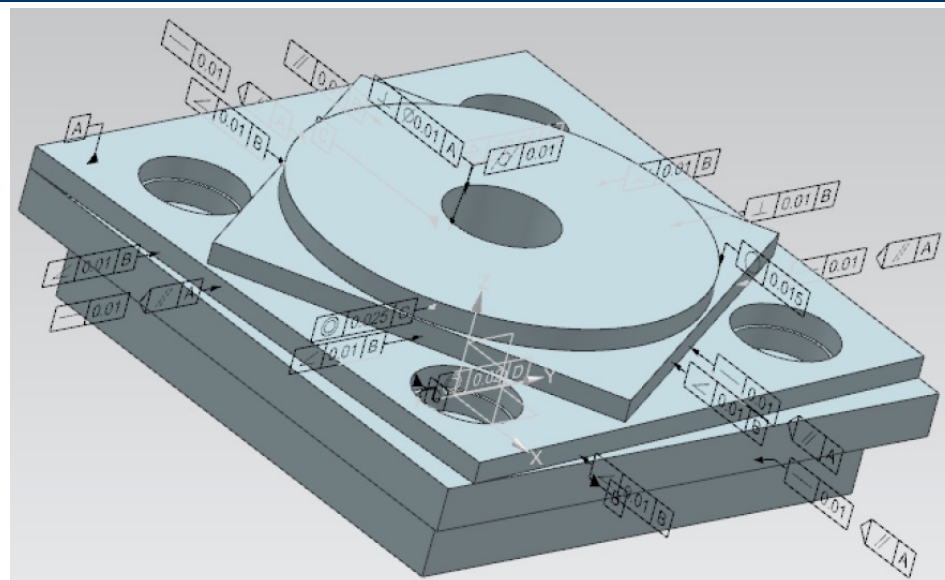
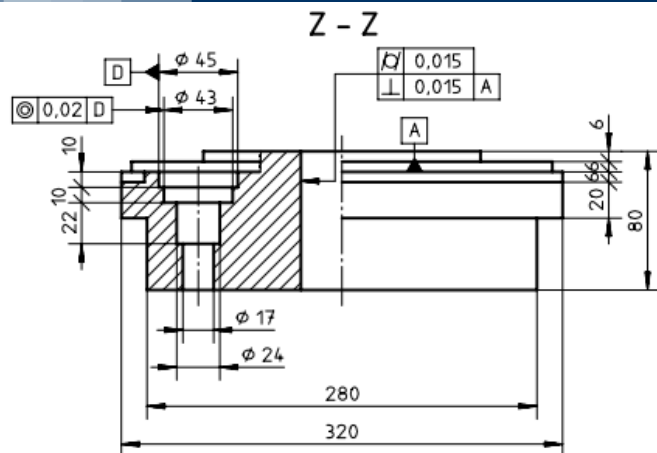
Envelope
(size controls the form)

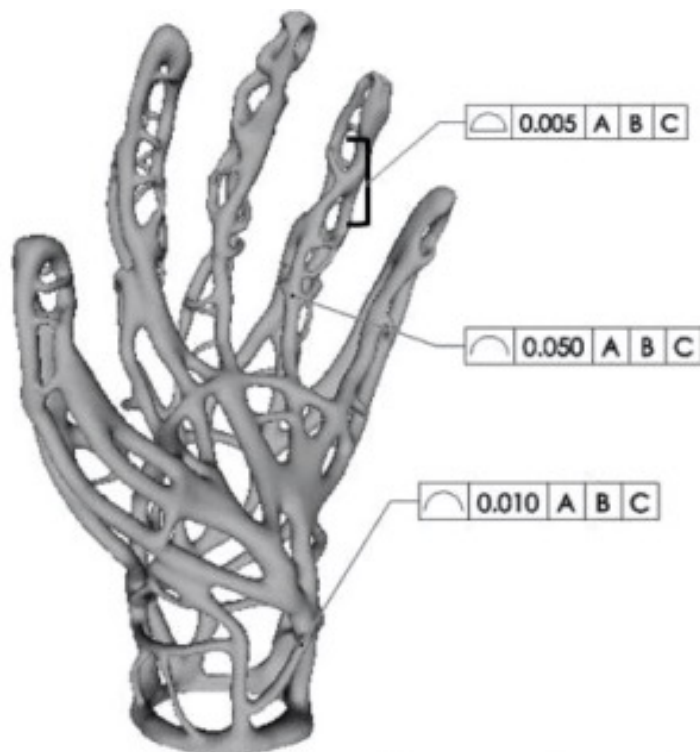
ISO



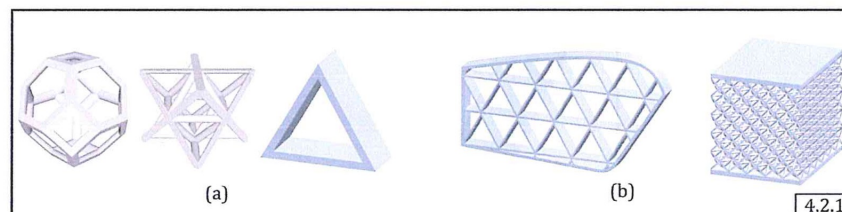
ASME





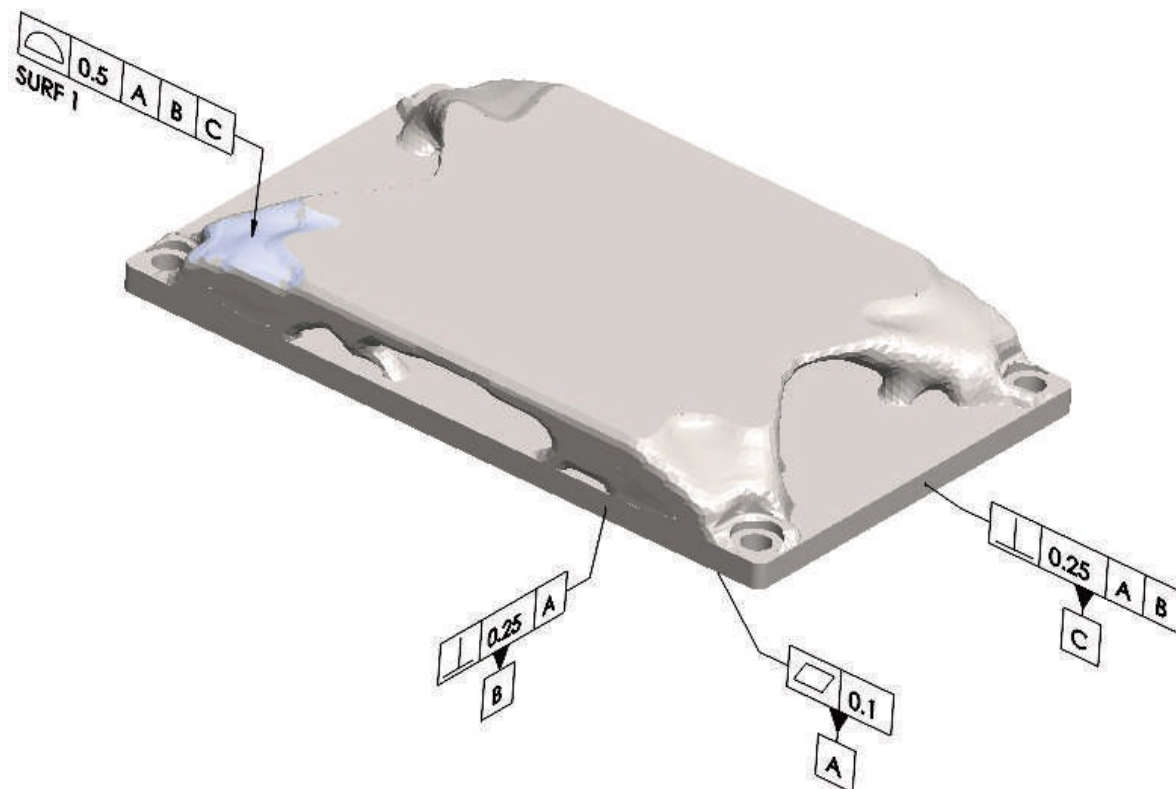


Lattice Structures (ISO/ASTM 52900):
Three dimensional geometrical arrangement composed of connective links between vertices (points) creating a functional structure.



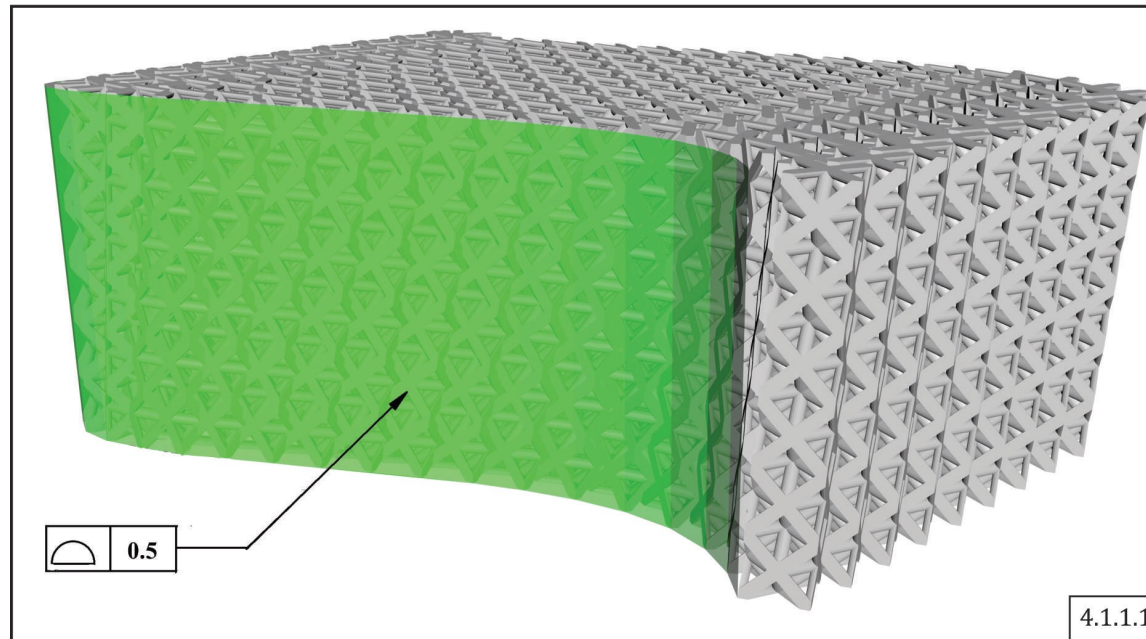
- a) Individual unit cell geometries
- b) Lattice structures

ASME Y14.46 - 2022



Bounded surface region

ASME Y14.46 - 2022



Nonplanar theoretical supplemental surface